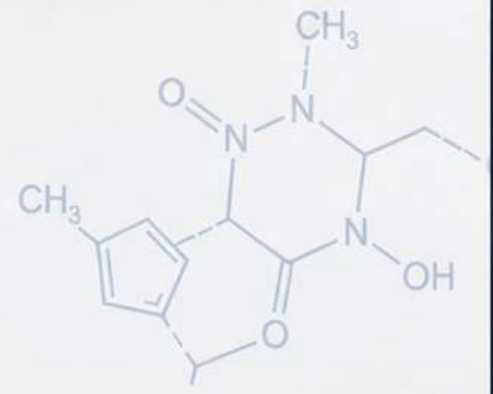




Lecture Notes

— on —

PHARMACOLOGY

DRUGS, MECHANISMS & THERAPEUTICS

2nd TERM

Lecture Classes by

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NOTES COLLECTED FROM

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*Knowledge is the
best medicine.*



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PHARMACOLOGY • 2ND TERM

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Note: In the lecture notes sequence, Topic 10 (*Antidepressants*) appears on pages 46–52, preceding Topic 08 (*Anti-Parkinson Drugs*, pages 53–56).

Quick guide • Short topic names for fast revision

LECTURE 1 (L-1)

CENTRAL NERVOUS SYSTEM & GENERAL DRUG CLASSIFICATIONS

Disease / Categories:

1. Central Nervous System Disorders:
 - i. Depression (**anti depressant drug**)
 - ii. Psychosis (**anti psychotic drug**)
 - iii. Epilepsy (**anti epileptic drug**)
 - iv. Parkinson (**anti parkinsonian drug**)
2. Sedative: **Benzodiazepine** and **barbiturate**
3. Anaesthetic agent:
 - i. General anaesthetic agent
 - ii. Local anaesthetic agent
4. Anti histamin (**anti histamin**)
5. Autacoids, eicosanoids
6. Steroid / glucocorticoids (**glucocorticoids**)
7. Analgesia / analgesics: **Opioid analgesics** and **non-opioid analgesics** / **NSAIDs** (**morphine**)
8. Antibiotics:
 - i. **Penicillin** — Alexander Flemming - 1929
 - ii. **Cephalosporin**
 - iii. **Macrolids** — **azithromycin** , **erythromycin** , **Clarithromycin**
 - iv. **Tetracycline**
 - v. **Aminoglycoside**
 - vi. **Nilezolid** (**Linezolid**)
 - vii. **Sulphonamide** — **cotrimoxazole**
 - viii. **Fluroquinolone** — **Ciprofloxacin** , **Rivofloxacin**
9. Anti tubercular drug
10. Anti malarial drug
11. Anti fungal drug
12. Anti amebic drug
13. Anti helminthic drug
14. Drugs used in Kala-azar
15. Cancer chemotherapy: (**anti cancer drug**)

SPECIAL DRUG CONCEPTS

- i. Essential drug
 - ii. P drug
 - iii. Rational and irrational prescribing
 - iv. Polypharmacy
-

PEPTIC ULCER DISEASE (PUD)

Common site: 1st part of duodenum and pyloric antrum.

DEFINITION

Ulceration of gastric mucosa due to acid-pepsin digestion is called peptic ulcer disease.

It is caused by imbalance between:

- i. Acid and pepsin
- ii. Mucosal defense
- iii. Helicobacter pylori

TREATMENT APPROACHES

- i. **Acid suppression by** →
 - H_2 receptor blockers
 - Proton pump inhibitor
 - Anti muscarinic drug
- ii. **Neutralization of acid and pepsin activity** → **Antacid**
- iii. **Eradication of H. pylori**
- iv. **Cyto-protection**

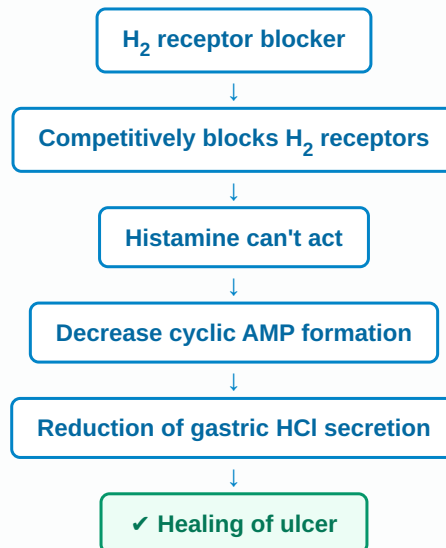
DRUGS USED FOR PUD

- i. H_2 receptor blocker: **Ranitidine** (Banned), **Cimetidine** (discontinued), **Famotidine**, **Nizatidine**
- ii. Proton pump inhibitor: **Omeprazole**, **Lansoprazole**, **Pantoprazole**, **Esomeprazole**, **Rabeprazole**
- iii. Anti muscarinic drug: **Pirenzepine**, **Propantheline** (Propathelene)
- iv. Antacid: **Aluminium hydroxide**, **magnesium trisilicate**, **calcium hydroxide**
- v. Cytoprotective agents: **Sucralfate**, Prostaglandin E_1 → **misoprostol**, **bismuth compound**
- vi. Antibiotics: **Amoxicillin**, **Clarithromycin**, **metronidazole**

H_2 RECEPTOR BLOCKERS

Action: Reduce basal and food stimulated HCl secretion.

MECHANISM OF ACTION



✓ INDICATIONS

- i. Peptic ulcer disease: gastric ulcer + Duodenal ulcer
- ii. Reflux oesophagitis
- iii. Prevention of Stress ulcer
- iv. Pre-anaesthetic medication
- v. Zollinger-ellision's syndrome
- vi. Extensive burn (to prevent curling ulcer)
- vii. Head injury (to prevent cushing ulcer)
- viii. Systemic mastocytosis (hypersecretory condition)

⚠ ADVERSE EFFECTS

- i. Headache
- ii. Dizziness
- iii. Myalgia
- iv. Diarrhoea
- v. Constipation
- vi. Done by **Cimetidine** :
 - Gynaecomastia → in male
 - Impotence → in male
 - Galactorrhea → in female

DIFFERENCE BETWEEN RANITIDINE AND CIMETIDINE

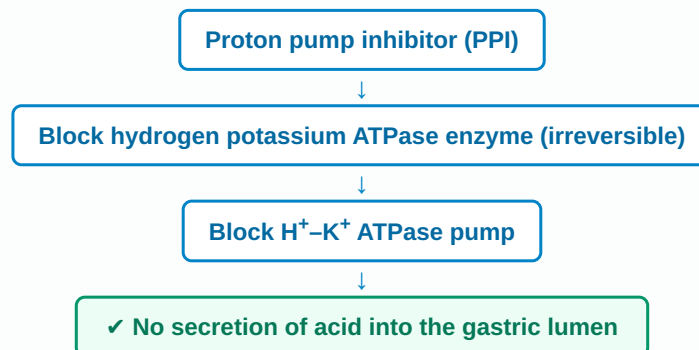
TRAITS	RANITIDINE (RANITIDINE)	CIMETIDINE (CIMETIDINE)
Potency	10 times more potent than Cimetidine	Less potent
Dose	150 mg BD (two times a day)	400 mg BD
Anti androgen effect	—	Block androgen receptor so cause gynaecomastia and impotency .
Acts on cytochrome enzyme	—	Inhibits cyto P-450 oxidase enzyme. So inhibit other drugs such as — Theophylline , Phenytoin , propranolol , anticoagulant .

PROTON PUMP INHIBITOR (PPI)

KEY PROPERTIES

- These are prodrug, especially **omeprazole** .
- Activated in acidic pH, pH < 3.
- Inhibit both basal and food stimulated acid secretion.

MECHANISM OF ACTION OF OMEPRAZOLE



✓ INDICATIONS

- Peptic ulcer disease
- Active duodenal ulcer
- Zollinger-Ellison's syndrome (tumor of pancreas)
- Gastro-oesophageal reflux disease
- Systemic mastocytosis

⚠ ADVERSE EFFECTS

- Nausea, vomiting, diarrhoea
- Headache
- Carcinoid tumor (prolong use)

For external (viva) →

- Prolonged use (vit B12 absorption ↓)
- Carcinoid syndrome
- osteoporosis & Bone fracture
- Hypochlorhydria

ANTACIDS

Antacid reacts with gastric HCl to produce salt and water.

Thus remove HCl and raise the stomach pH and inhibit pepsin activity. So relief of heart burn, indigestion.

NEUTRALIZATION REACTIONS

ex. Aluminium Hydroxide $[\text{Al}(\text{OH})_3] + \text{HCl} \rightarrow \text{AlCl}_3 + \text{H}_2\text{O}$

Sodium bicarbonate + HCl $\rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$

CLASSIFICATION OF ANTACIDS

1. Systemic antacid (Absorbable):

Water soluble and completely absorbed in systemic circulation and produce metabolic alkalosis.

ex. Sodium bicarbonate, Potassium bicarbonate.

⚠ WARNING: Sodium containing antacid dangerous in cardiac and renal disease.

2. Non systemic antacid (Non absorbable):

Water insoluble and not absorbed in systemic circulation.

ex. $\text{Al}(\text{OH})_3$, $\text{Mg}(\text{OH})_2$, Mg trisilicate, Mg Carbonate.

CHEMICAL COMPOSITION OF ANTACIDS

Chemically, antacid consists of **weak basic part** and **metallic cation part**.

COMPONENT	FORM / COMPOSITION	PHARMACOLOGICAL ROLE
Weak basic part	In the form of hydroxide, oxide, bicarbonate, trisilicate	Responsible for neutralization of acid.
Metallic cation part	e.g., Aluminium, Magnesium, Calcium	Responsible for pharmacodynamic property and adverse effect.

PROPERTIES OF ANTACID (IDEAL PROPERTIES)

- Should be palatable, non-irritant, non-toxic
- Should be cheap and available
- Rapid onset of action
- Should not interfere with absorption of food

✓ INDICATIONS OF ANTACIDS

- Hyperacidity
- Gastric ulcer
- Duodenal ulcer
- Gastritis
- Reflex esophagitis *GERD*
- Non ulcer dyspepsia

⚠ **CATION-SPECIFIC ADVERSE EFFECTS & CLINICAL NOTES**

- **Aluminium (Al)** cause **constipation**.
- **Magnesium (Mg)** cause **diarrhoea**.
- **Calcium (Ca)** cause **constipation** and **milk alkali syndrome**.
- **All antacids cause acid rebound specially Ca Carbonate (Calcium Carbonate)**.

Bismuth compound → Cytoprotection

GLUCOCORTICOID: CLASSIFICATION & PHARMACOKINETICS

I. ACCORDING TO THE DURATION OF ACTION

1. Short acting ($t^{1/2} < 12$ hours)

ex. hydrocortisone (cortisol)

2. Intermediate acting ($t^{1/2} = 12-36$ hours)

ex. prednisolone , methylprednisolone , triamcinolone , Deflazacort .

3. Long acting ($t^{1/2} > 36$ hours)

ex. Dexamethasone , Betamethasone .

II. ACCORDING TO SOURCE

1. Natural — hydrocortisone (cortisol)

2. Synthetic —

- Short acting (Hydrocortisone) — IV
- Intermediate acting — prednisolone , methylprednisolone , triamcinolone , Deflazacort .
- Long acting — Dexamethasone , Betamethasone .

III. ACCORDING TO ROUTE OF ADMINISTRATION

• Oral:

- Prednisolone → IV, IM
- methylprednisolone → IV, IM
- Dexamethasone

• Parenteral:

- Prednisolone → Slow I.V, IM
- methylprednisolone → Slow I.V, IM
- Triamcinolone — IM, Intraarticular
- Hydrocortisone — IV
- Dexamethasone — IV, IM

• Inhalational:

Fluticasone , Beclomethasone , Budesonide , mometasone .

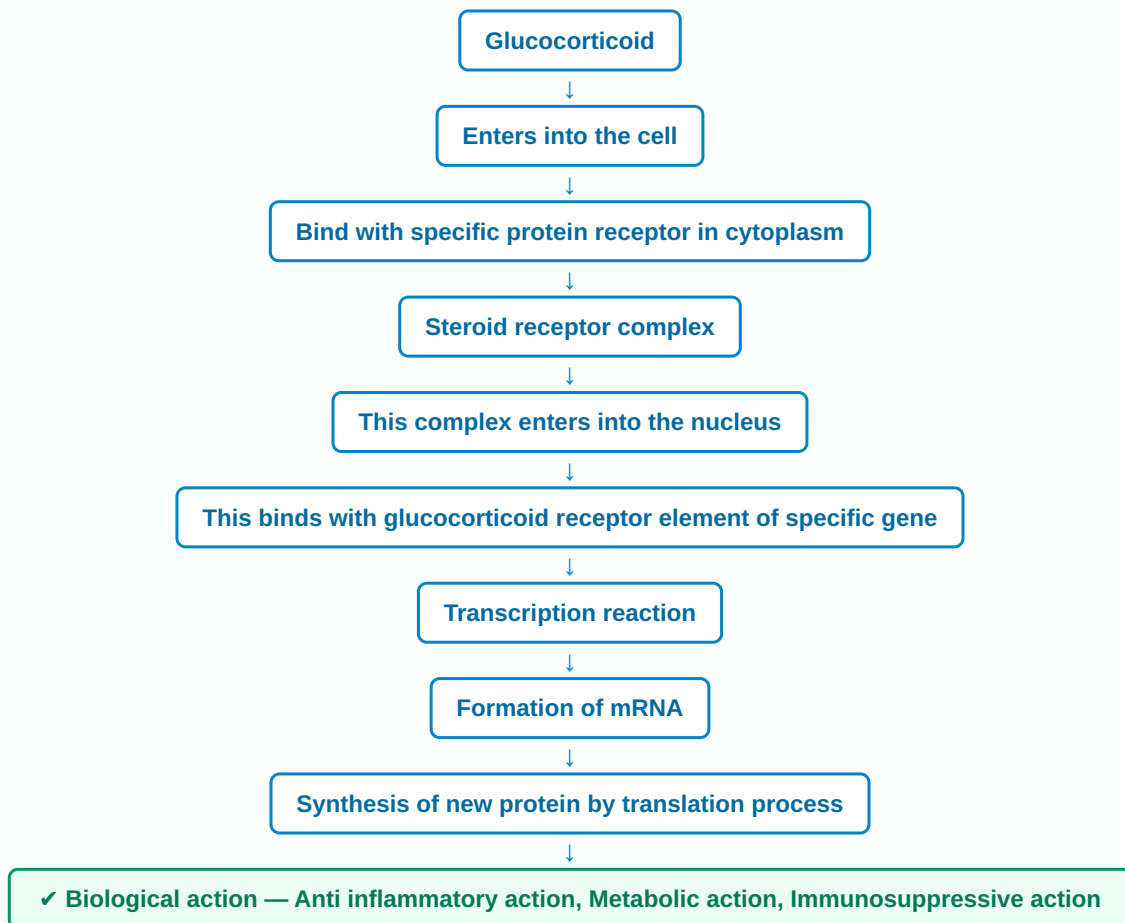
• Topical:

Hydrocortisone , clobetasol , mometasone , fluticasone , flunisolide .

IV. ACCORDING TO GLUCOCORTICOID ACTIVITY / POTENCY

- Hydrocortisone (cortisol) → 1 (less potent)
 - Prednisolone , methylprednisolone → 5 (moderate potent)
 - Triamcinolone → 5 (moderate potent)
 - Dexamethasone → 30 (more potent)
 - Betamethasone → (25-30) more potent
-

MECHANISM OF ACTION OF GLUCOCORTICOIDS



ACTION OF GLUCOCORTICOID / STEROID

1. Anti inflammatory action
2. Metabolic action
3. Immunosuppressive action
4. Permissive action

1. ANTI INFLAMMATORY ACTION

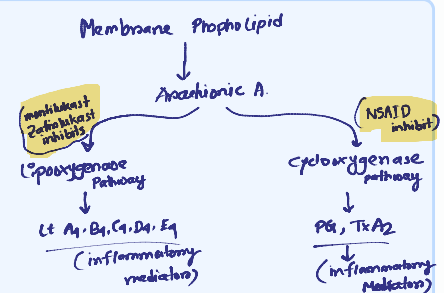
PATHWAY:

Membrane phospholipid (cell membrane)

↓ [Phospholipase A₂ ← Inhibited by Steroid]

Arachidonic acid

- → **Lipoxygenase:** Leukotriene A₄, B₄, C₄, D₄, E₄ (Inflammatory mediator)
- → **Cyclooxygenase:** Prostaglandin, Thromboxane (Inflammatory mediator)



2. METABOLIC ACTION

- **On Carbohydrate metabolism:**
Increase blood glucose level by —
 - i. Stimulating gluconeogenesis
 - ii. Inhibit glucose uptake.

- **On protein metabolism:**

- i. Enhance protein catabolism

So prolong steroid therapy results in:

1. Muscle wasting
2. Osteoporosis
3. Removal of skin protein
4. Skin becomes thin → easy bruise
5. Increase capillary permeability → easy bruise and striae.

- **On fat metabolism:**

- i. Enhance lipolysis.

3. IMMUNOSUPPRESSIVE ACTION

1. Inhibit formation of antibody.
2. Decrease cell mediated immunity.
3. Inhibit release of interleukin-1 from macrophage and interleukin-2 from T-lymphocyte.
4. Inhibit activity of complement.
5. Limit the ability of macrophage to phagocytosis and kill micro-organism.

4. PERMISSIVE ACTION

DEFINITION

It is a physiological phenomenon where presence of one hormone is required for another hormone to exert its full effect on a target cell or organ.

1. Adrenaline can't fully stimulate contraction of vascular smooth muscle or cause bronchodilation in absence of cortisol or glucocorticoid.
2. Glucocorticoid is required to stimulate gluconeogenesis.
3. Glucocorticoid required for catecholamines, growth hormone, adrenocorticotrophic hormone to stimulate breakdown.

INDICATION OF STEROID

1. Replacement therapy (endocrine use)
2. Non replacement therapy (non endocrine use)
3. Other indications

✓ REPLACEMENT THERAPY (ENDOCRINE USE)

1. Acute adrenal insufficiency (addisonian's crisis)
2. Chronic adrenal insufficiency
3. Congenital adrenal hyperplasia
4. Diagnosis of cushing syndrome by **dexamethasone** suppression test.



✓ NON REPLACEMENT THERAPY (NON ENDOCRINE USE)

1. Lung:

- Bronchial asthma
- Aspiration pneumonia
- Pulmonary edema
- Fetal lung maturity
- Respiratory distress syndrome (RDS)

2. Allergic reaction:

- Anaphylactic shock
- Angioneurotic edema
- Urticaria
- Serum sickness.

✓ OTHER INDICATIONS

1. Arthritis:

- Rheumatoid Arthritis
- Osteoarthritis
- Gout

2. Collagen vascular disease:

- SLE (Systemic Lupus Erythematosus)
- Polyarteritis nodosa

3. Kidney:

- nephrotic syndrome
- Glomerulonephritis

4. CNS:

- cerebral edema
- meningitis

5. Skin:

- eczema
- pemphigus
- Exfoliating dermatitis
- Steven's – Johnson's syndrome

6. CVS:

- Rheumatic carditis

7. Eye:

- Allergic conjunctivitis
- Optic neuritis
- Uvitis
- Retinitis

8. GIT:

- Ulcerative colitis
- Irritable bowel syndrome
- Tropical sprue
- Crohn's disease
- Coeliac disease

9. Blood:

- ALL : Acute Lymphoblastic Leukemia
- Idiopathic thrombocytopenic Purpura
- multiple myeloma.

10. Thyroid gland:

- Hodgekin's Lymphoma
- Thyroiditis

11. Organ transplantation and skin grafting:

- To suppress immunity

✗ CONTRAINDICATIONS OF GLUCOCORTICOIDS

1. peptic ulcer disease (PUD)
2. Hypertension
3. CCF (congestive cardiac failure)
4. Glaucoma
5. TB (Tuberculosis)
6. Osteoporosis
7. DM (Diabetes Mellitus)
8. Psychosis
9. epilepsy
10. Renal failure
11. Keratitis caused by Herpes simplex virus
12. Viral and fungal infection

⚠ ADVERSE EFFECTS OF GLUCOCORTICOIDS

1. Adrenal and pituitary suppression
2. **CVS** → increase BP
3. **GIT** →
 - Peptic ulceration
 - Pancreatitis
4. **Bone and muscle** →
 - osteoporosis
 - muscle wasting
 - pathological fracture
5. **Endocrine** →
 - weight gain
 - Increase blood glucose level.
6. **Skin** →
 - Thin
 - Easy bruise
7. **Eye** →
 - Posterior subcapsular cataract
 - Glaucoma
8. Delayed wound healing, fungal infection
9. **CNS** →
 - Depression
 - Euphoria
 - Psychosis
 - Insomnia
10. **Fetus** → Cleft palate

⚠ WITHDRAWAL SYNDROME (TAPERING DOSE)

1. fever
2. Arthralgia
3. myalgia
4. rhinitis
5. Adrenocortical insufficiency

☐ Precaution of steroid

- Blood pressure check
- X-ray (bone density)
- Blood sugar check

SEDATIVE — HYPNOTIC

DEFINITIONS & TERMINOLOGY

- **Sedative:** The agents that produce calming or quietening effect, reduce excitement and may induce drowsiness.
- **Hypnotic:** The agent that induces sleep, resembling natural sleep.
- **Sedation:** It is a state of drowsiness, calmness, quietness without producing actual state.
- **Hypnosis:** It is a state of CNS depression, resembling natural sleep.
- ~~Anxiolytic~~ / **Anxiolytic:** Agent / drugs that reduce anxiety. State on condition of anxiety. → *Anxiolysis*

CLASSIFICATION OF SEDATIVE / HYPNOTIC

1. Benzodiazepine:

- i. Short acting: **midazolam**, **triazolam**.
- ii. Intermediate acting: **lorazepam**, **oxazepam**, **temazepam**, **alprazolam**.
- iii. Long acting: **Diazepam**, **nitrazepam**, **clonazepam**, **flurazepam**.

2. Non-Benzodiazepine:

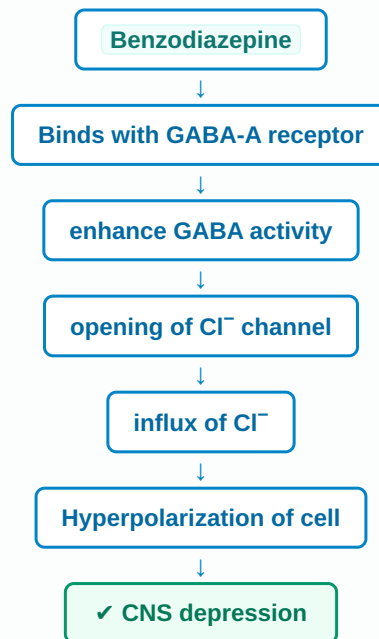
zolpidem, **zaleplon**, **zopiclone**, **buspirone**.

3. Barbiturate:

- i. Ultra short acting: **thiopental sodium** (anaesthetic)
- ii. Short acting: **Pentobarbitone**, **Secobarbitone**, **Hexobarbital**.
- iii. Intermediate acting: **butobarbitone**, **Amitobarbitone**.
- iv. Long acting: **Phenobarbitone**.

BENZODIAZEPINE *→ must learn + Post*

MECHANISM OF ACTION



CNS EFFECTS

- i. Sedation
- ii. Hypnosis
- iii. Anxiolysis
- iv. Anaesthesia
- v. Muscle relaxation
- vi. Anti-convulsive effect
- vii. Amnesia

✓ INDICATION / USE OF DIAZEPAM

- i. Insomnia
- ii. Anxiety
- iii. Convulsive disorder — tetanus, epilepsy, eclampsia.
- iv. As muscle relaxant.
- v. As preanaesthetic medication.
- vi. During alcohol withdrawal.

✗ CONTRAINDICATION (Diazepam)

- myasthenia gravis.

- Pregnancy and lactating mothers
 - Ataxia
 - Acute narrow angle glaucoma
 - Severe hepatic and renal dysfunction
- Bristy maam*

PHARMACOLOGICAL EFFECTS OF BENZODIAZEPINE

Part of Benzodiazepine "Sedative Hypnotic"

ACTIONS & EFFECTS

1. **Anxiolytic effect:** It produces sedation, reduces aggressiveness & exerts calming effects by lowest effective dose. (5mg, also works in 2.5mg).
2. **Hypnotic effect:** It produces drowsiness & encourages the onset and maintenance of sleep (produced by higher therapeutic dose). BDZs preserves normal sleep pattern.
3. **Muscle relaxant effects:** It inhibits polysynaptic reflexes and internuncial (interneuron connecting afferent & efferent nerve) transmission at the level of the spinal cord.
At higher dose it may depress transmission at the skeletal neuromuscular junction. This precipitates anti-convulsant effect at higher dose.
Diazepam & **Clonazepam** has more marked muscle relaxant properties.
4. **Anti convulsant effect:** By inhibition of development & spread of epileptiform electrical activity in the CNS without marked CNS depression.
Diazepam , **clonazepam** , **Lorazepam** has more marked anti convulsant effect.
5. **Anesthetic effect:** At high dose of certain BDZ depress the CNS to the point as stage III general anesthetic.
Diazepam , **Midazolam** , **Lorazepam** are used in anesthesia with other drugs.
In high dose, it may cause post anesthetic respiratory depression.
6. **Effects on respiration and circulation:** Up to hypnotic dose, no significant effects. But in high dose, produce depression of the vital centers in medulla.

⚠ ADVERSE EFFECTS (Diazepam)

- i. Dry mouth
- ii. Light headache
- iii. Lassitude
- iv. Ataxia
- v. Motor in coordination
- vi. Drowsiness
- vii. Confusion
- viii. Impaired driving skill
- ix. Tolerance
- x. Dependency

+ (xi) Altered libido
(xii) Anterograde Amnesia } Brisky moam

ANTAGONIST

Flumazenil .

LECTURE 6 (L-6)

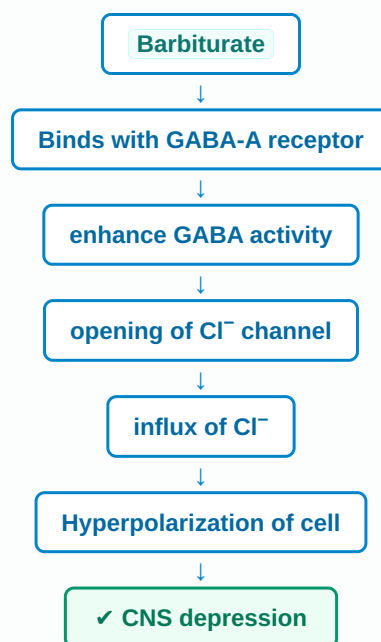
BARBITURATE

CLASSIFICATION

- i. Ultra short acting: Thiopental sodium (anaesthetic)
- ii. Short acting: Pentobarbitone , Secobarbitone , Hexobarbital .
- iii. Intermediate acting: butobarbitone , amobarbitone .
- iv. Long acting: Phenobarbitone .

MECHANISM OF ACTION

(Same as Benzodiazepine)



PHARMACOLOGICAL ACTION

- **CNS:**

- i. Sedation
- ii. Hypnosis
- iii. anti convulsion
- iv. anaesthesia

- **Respiratory:** Respiratory depression

- **CVS:** decrease Heart rate, decrease blood pressure.

- **Kidney:** Oliguria

- **Eye:** myosis

✓ INDICATIONS

- i. Anaesthesia
- ii. convulsion
- iii. Epilepsy
- iv. Pre-anaesthetic medication
- v. kernicterus

⚠ ADVERSE EFFECTS

- i. Drowsiness
- ii. Hallucination
- iii. Hangover
- iv. Addiction
- v. Tolerance
- vi. Dependence
- vii. Drug automatism
- viii. Idiosyncrasy

DIFFERENCE BETWEEN BENZODIAZEPINE AND BARBITURATE (D/B)

BENZODIAZEPINE	BARBITURATE
High therapeutic index	Low therapeutic index
Cause less pronounced CNS depression	Cause more pronounced CNS depression.
Does not induced hepatic microsomal enzyme	induces
cause less respiratory depression	more
does not produce hangover effect	produces
withdrawal syndrome — less	more

1st Stages of CNS depression:

Sedation → Hypnosis → Narcosis → Anaesthesia → Coma → Death

AUTACOIDS (LOCAL HORMONE)

DEFINITION

Autacoids are the chemical substances that are produced locally and act at very close to the origin.

📌 Name of the autacoids:

1. **Biogenic amines** — histamine, serotonin.

2. **Lipid derived** — eicosanoids:

- Prostaglandin
- Leukotriens
- Thromboxane A₂

3. **Endogenous peptide** — Bradykinin, Angiotensin-2

HISTAMINE / HISTOSIS

GENERAL CONCEPT

It means tissue amine. It is a major media for allergic and inflammatory process.

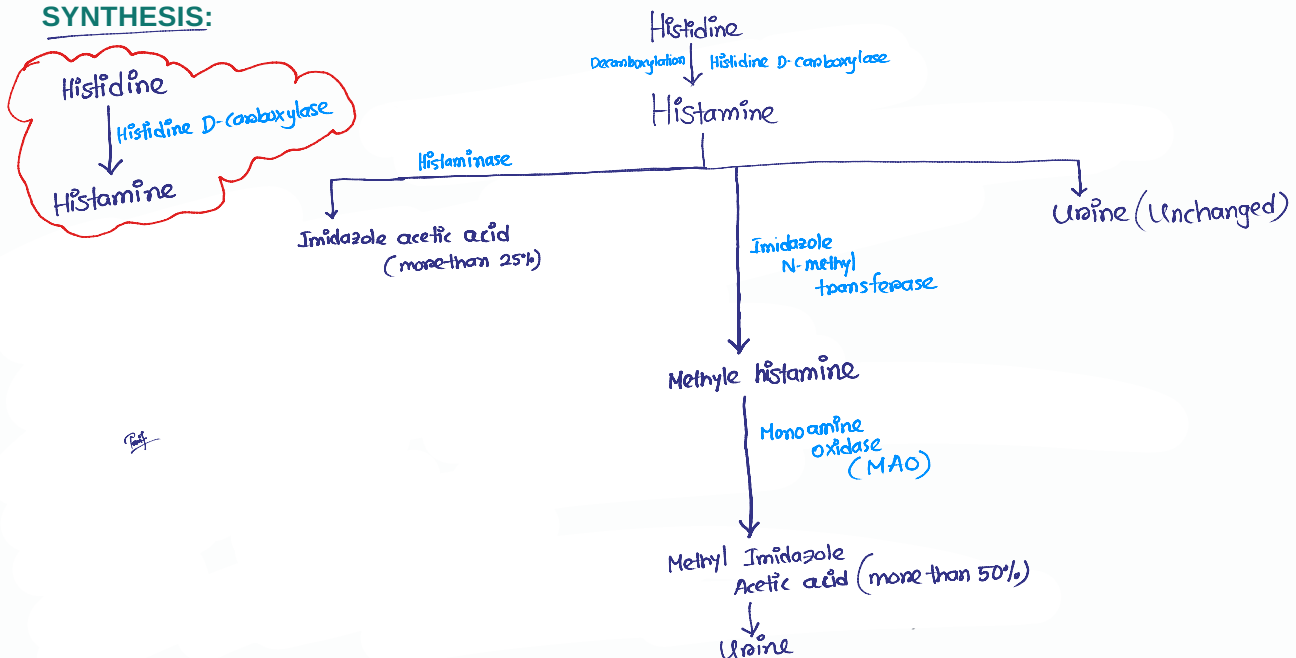
SITE OF HISTAMINE RELEASE

1. **Mast cell site:**
pulmonary tissue, skin, nose, mouth, feet, intestinal mucosa.
2. **Non-mast cell site:**
Neuron, ECL (enterochromaffin cell), Basophil.

Site of Storage of histamine — mast cell, basophil. (chief storage site)

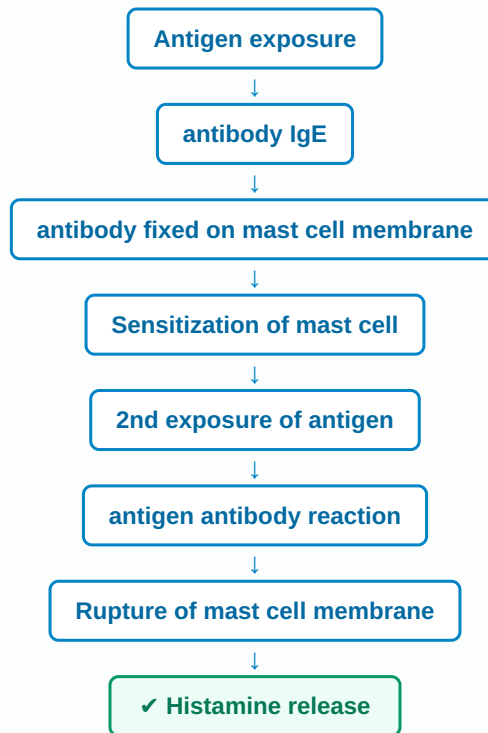
SYNTHESIS AND METABOLISM OF HISTAMINE

SYNTHESIS:

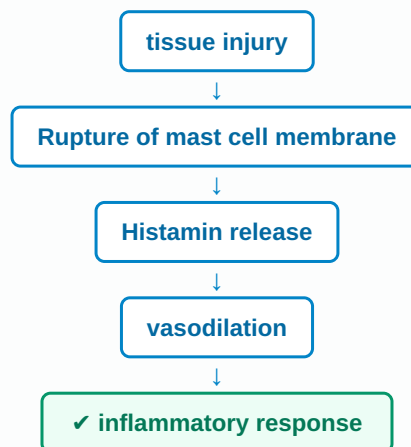


HISTAMINE RELEASE MECHANISMS

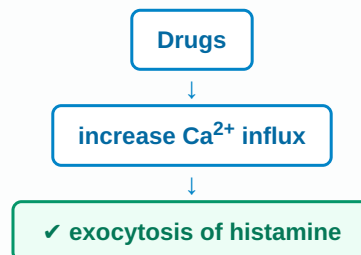
1. IMMUNOLOGICAL RELEASE



2. MECHANICAL RELEASE



3. CHEMICAL RELEASE / DRUG INDUCED



DRUG CAUSING HISTAMINE RELEASE:

D-tubocurarine, Suxamethonium, Codeine, morphine, Penicillin, radiocontrast media.

HISTAMINE RELEASE INHIBITOR:

(Act by decreasing in Ca^{2+} influx, so decrease histamine release)

1. **Histamine** (How → H_3 receptor act by decreasing in Ca^{2+} influx, so decrease histamine release)
2. **Glucocorticoid**
3. **β -agonist**
4. **Aminophylline**
5. **Chromolyn Sodium**

HISTAMINE RECEPTORS

1. H_1 RECEPTOR

- **Site** — Bronchus, adrenal medulla, skin, blood vessel, GIT, urinary bladder, thalamus, hypothalamus, cerebellum.
- **Action** — H_1 -receptor act by **increasing intracellular Ca^{2+}** .

2. H_2 RECEPTOR

- **Location** — Gastric parietal cell, heart, CNS.
- **Action** — by **increasing intracellular cAMP**.

3. H_3 RECEPTOR

- **Location** — Presynaptic, myenteric plexus.
- **Action** — Act by **decrease in Ca^{2+} influx**, so **decrease histamine release**.

ROLE OF HISTAMINE IN ALLERGY AND ANAPHYLAXIS

1. Contraction of smooth muscle
2. Dilation of capillary and increase capillary permeability.
3. Stimulation of secretion
4. Stimulation of sensory nerve ending.

LECTURE 8 (L-8)

PHARMACOLOGICAL ACTION OF HISTAMINE

1. **Lung:** Bronchoconstriction, increased bronchial secretion.
2. **GIT** → Contraction of muscle → abdominal cramp
 - i. Parietal cell → Increased HCl secretion
 - ii. Increase pepsin and IF secretion.
3. **On blood vessels:**
 1. Relaxation of arteriole and pre capillary sphincter → so cause vasodilation.
 2. Contraction of post capillary venule → increase capillary permeability.
4. **On CVS:**
 1. Vasodilation → so decrease blood pressure
 2. Increase Heart rate and force of contraction → cause secondary hypertension (due to release of adrenaline from adrenal medulla H_1 receptor)

5. Skin —

1. Itching and pain
2. Cause **lewis-h-triple response**:
 - a. Redness (due to capillary dilatation)
 - b. Wheal (due to capillary permeability)
 - c. Flare (due to arteriolar dilatation and axon reflex)

6. Adrenal medulla:

Increase BP (due to release of adrenaline)

7. CNS:

1. acts as neurotransmitter
2. Regulate drinking, ADH release, temperature and pain.

PHYSIOLOGICAL ROLE OF HISTAMINE:

1. Plays role in gastric HCl secretion.
2. Endogenous histamine plays role in immediate hypersensitivity and allergic reaction.
3. Regulation of micro-circulation (important for inflammation)
4. Acts as neurotransmitter in CNS.

✓ INDICATION OF HISTAMINE

1. Diagnosis of Pernicious anemia.
2. Diagnosis of pheochromocytoma.
3. Diagnosis of the integrity of the reflex arc.
4. pulmonary function test.

✗ CONTRAINDICATIONS OF HISTAMINE

1. Bronchial asthma
2. Peptic ulcer disease
3. Hypotension

⚠ ADVERSE EFFECTS OF HISTAMINE

1. Bronchospasm
2. Flashing
3. Itching
4. Headache (intense)
5. Abdominal cramp
6. Diarrhoea
7. Hypotension

ANTI HISTAMINE

MECHANISM / DEFINITION

It blocks the H₁ receptor or antagonise the action of histamine.

CLASSIFICATION:

A. 1st generation (Sedative)

1. **Highly sedative:** (*in case of baby*)

eg. **Diphenhydramine**, **Dimenhydrinate**, **Hydroxyzine**, **promethazine**

2. **Moderately sedative:**

eg. **pheniramine**, **meclizine**, **Cyclizine**, **cinnarizine**, **Cyproheptadine**.

3. **Mild sedative:**

eg. **chlorpheniramine**, **Dexchlorpheniramine**, **clemastine**, **tripolidine** (less used)

B. 2nd generation (Non selective):

eg. **Fexofenadine**, **Cetirizine**, **Levocetirizine**, **Loratadine**, **desloratadine**, **Rupatadine**, **azelastine**, **mizolastine**, **ebastine**.

✓ INDICATIONS OF ANTI HISTAMINES

1. **Allergic reaction:** urticaria, rhinitis, conjunctivitis, anaphylactic shock, atopic dermatitis.
2. **Motion sickness:** **promethazine**, **diphenhydramine**
3. **Emesis:** **cyclizine**, **meclizine**.
4. **Parkinsonism**
5. **Pre anaesthetic medication**
6. **Sedation**

⚠ ADVERSE DRUG EFFECT

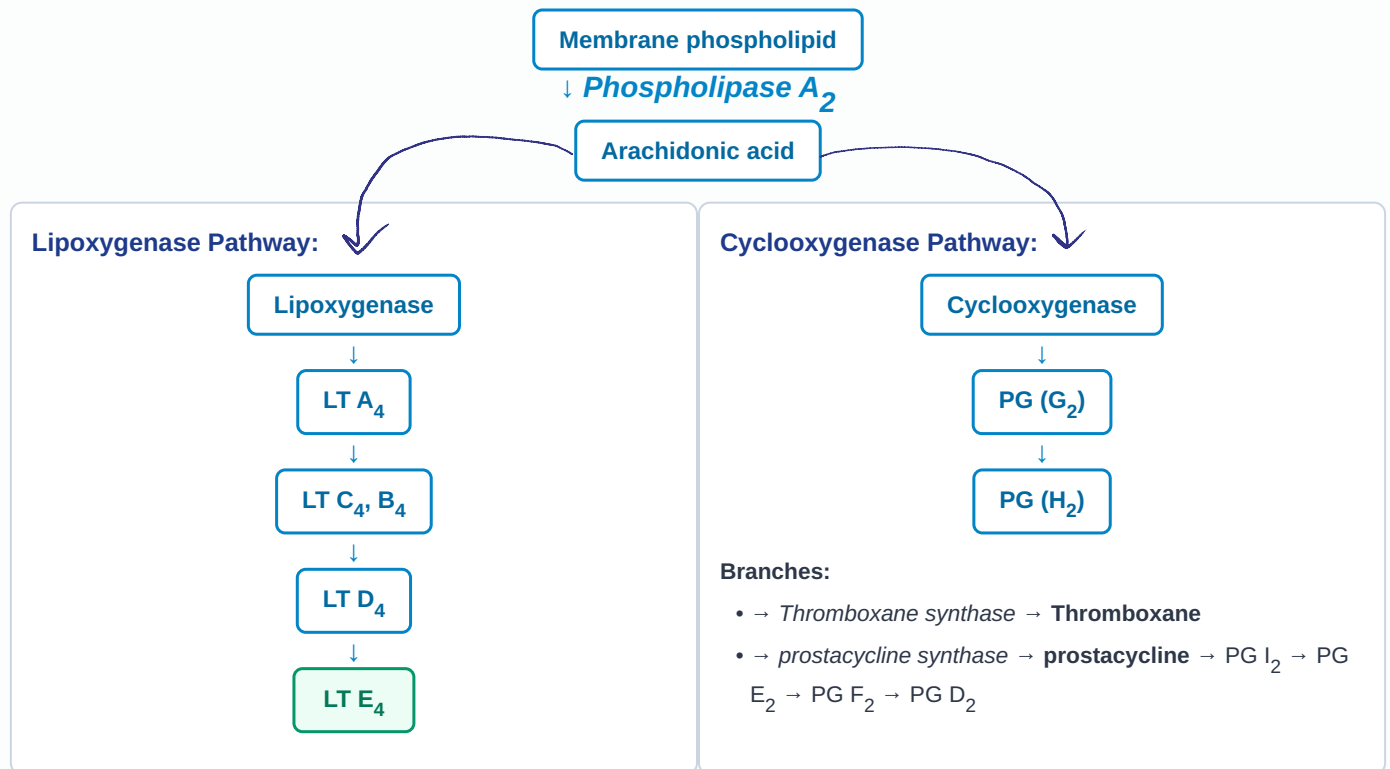
1. **CNS:** sedation, drowsiness, fatigue, inco-ordination, nervousness
2. **Ocular:** disturbance in vision.
3. **Anticholinergic action:** Dry mouth, constipation, urinary retention.
4. **GIT:** epigastric distress
5. **Agranulocytosis.**

COMPARISON OF ADVANTAGES

NON SELECTIVE	SEDATIVE
1. less lipid soluble	1. High lipid soluble
2. can't enter into CNS (No CNS depression)	2. Enters into CNS
3. Cause no sedation	3. Causes most sedation.
4. Long duration of action.	4. Short duration of action.

EICOSANOID

BIOSYNTHESIS OF EICOSANOID:



RESPIRATORY:

Bronchial asthma: **Histamine** and **antihistamine** , **Glucocorticoid** .

ANALGESIC: DEFINITIONS & CLASSIFICATION

DEFINITIONS

Analgesic: Drug that reduce pain due to multiple cause are called analgesic.

Pain: It is an unpleasant sensory and emotional experience associated with actual or potential tissue damage.

CLASSIFICATION OF ANALGESICS (2 TYPES):

1. **Opioid analgesic / Narcotic analgesic:**
ex. **morphine**, **Pethidine (Meperidine)**, **codeine**.
2. **Non-opioid / non narcotic analgesic (NSAIDs):**
ex. **Aspirin**, **Diclofenac**, **Naproxen**, **ketorolac**.

DIFFERENCE BETWEEN OPIOID / NARCOTIC AND NON-OPIOID / NON-NARCOTIC (D/B):

OPIOID / NARCOTIC	NON-OPIOID / NON NARCOTIC
(i) Highly Potent	(i) less potent
(ii) mainly acts on CNS and cause CNS depression (prototype — morphine)	(ii) mainly act peripherally (ex. aspirin)
(iii) Produce narcosis	(iii) don't produce narcosis.
(iv) Specific receptor mechanism	(iv) PG synthesis
(v) used in visceral pain	(v) Somatic pain
(vi) Low therapeutic index	(vi) high therapeutic index
(vii) No antiinflammatory, antiplatelet and antipyretic action	(vii) has actions
(viii) Cause — dependence, tolerance, addiction.	(viii) cause — non narcotic analgesic → GIT — irritation

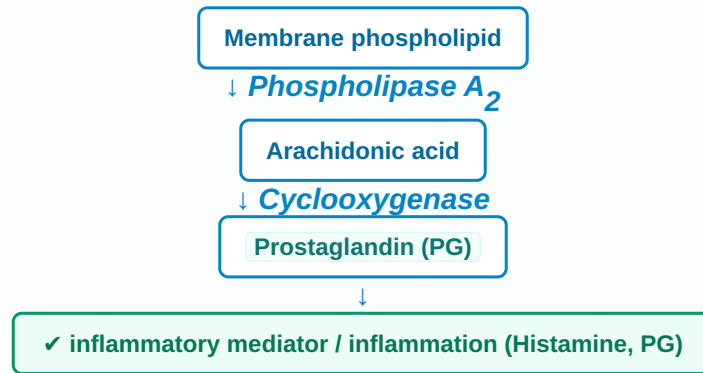
CHOICE OF ANALGESIC:

- (i) **Mild pain** — eg. non-opioid / NSAID — ex. **paracetamol**, **Ibuprofen**, **Aspirin (low dose)**, **ketorolac**.
- (ii) **Moderate pain** — low efficacy opioid — Like — **codeine**, **dihydrocodeine**.
- (iii) **Severe pain:** High efficacy opioid — ex. **morphine**, **pethidine**.
- (iv) **Acute pain:** opioid analgesic + sedative — ex. **morphine** / **pethidine** + **diazepam**.

CLASSIFICATION OF ANTI INFLAMMATORY DRUG:

1. **Steroidal anti inflammatory drug:** **glucocorticoid**
2. **Non steroidal anti inflammatory drug:** **Aspirin**, **ketorolac**

MECHANISM OF ACTION OF NSAID:



Extra (Bobby mom)

House keeping action of COX-1

- a. Stomach → Giving cytoprotection by
 - ① ↓ Gastric acid secretion
 - ② ↑ Gastric mucosal blood flow
 - ③ ↑ secretion of mucus and HCO_3^-
- b. Kidney → auto regulation of renal blood flow
- c. Platelet → Maintains the normal platelet flow
- d. Uterus → Inhibition of parturition

COMPLETE CLASSIFICATION OF NSAIDS

(1) ACCORDING TO SELECTIVITY

1. Non selective (Inhibit both COX-I and COX-II enzyme):

- a. Salicylic acid derivatives / salicylate — aspirin
- b. Fenamic acid derivatives / fenamate — ex. mephenamic acid , tolphenamic acid
- c. Enolic acid derivatives — Piroxicam
- d. Acetic acid derivatives — ketorolac , Indomethacin
- e. Pyrazolone derivatives — phenylbutazole
- f. Paraamino-phenol derivatives — paracetamol (acetaminophen)
- g. Propionic acid derivatives — Ibuprofen , ketoprofen , Phenoprofen etc.

2. selective
COX-II inhibitors:

ex. Celecoxib , Rofecoxib , etoricoxib , valdecoxib .

COX-II preferential inhibition (Inhibition of COX-II but weak inhibition of COX-I):

ex. meloxicam , Etodolac , Nimesulide .

(2) ACCORDING TO ANTI INFLAMMATORY ACTION

1. Weak anti inflammatory effect:

ex. paracetamol (acetaminophen)

2. Mild to moderate anti inflammatory effect:

- a. Propionic acid derivatives / propionates: Ibuprofen , ketoprofen , Phenoprofen , Flurbiprofen
- b. Fenamic acid derivatives / fenamates: eg. mephenamic acid , tolphenamic acid .
- c. Non acidic drug: eg. Nabumetone , Aceclofenac .

3. Strong anti inflammatory effect:

- a. Salicylic acid derivatives: eg. aspirin , Diflunisal
- b. Acetic acid derivatives: eg. ketorolac , Indomethacin , Nabumetone .
- c. Pyrazolone derivatives: eg. Phenylbutazole .
- d. Enolic acid derivatives: ex. Piroxicam , Tenoxicam .

(3) ACCORDING TO POTENCY AND EFFICACY

1. Low potency / efficacy: ex. paracetamol .

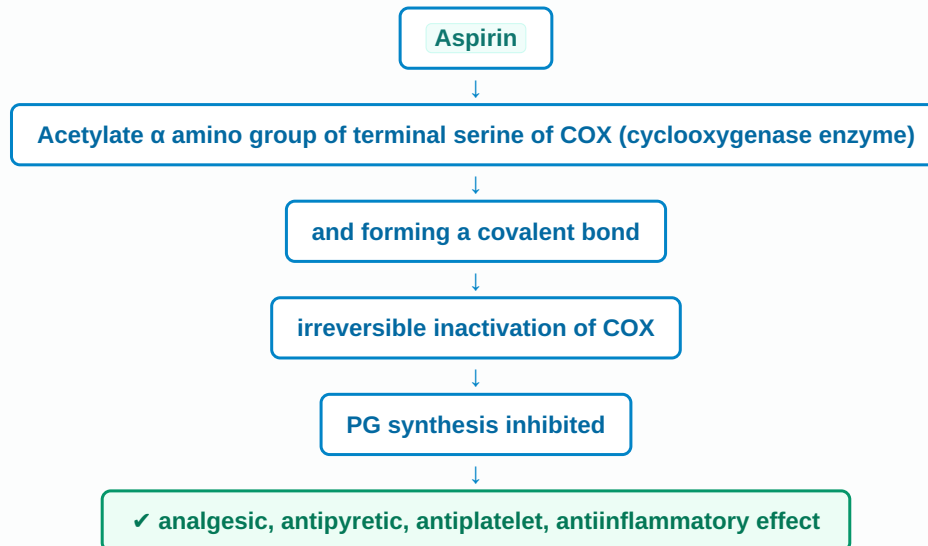
2. Moderate efficacy: ex. Ibuprofen , phenoprofen , mephenamic acid .

3. High Potency (Strong): ex. Aspirin , Diclofenac , Phenylbutazole , Piroxicam , Tenoxicam , Indomethacin .

ASPIRIN

Source: Willow Bark tree.

MECHANISM OF ACTION OF ASPIRIN:



PHARMACOLOGICAL ACTION OF ASPIRIN:

1. **Analgesic effect** (CNS and PNS effect)
2. **Anti Platelet**
3. **anti inflammatory**
4. **anti pyretic** (Peripheral effect, no CNS effect)

✓ INDICATION OF ASPIRIN

1. **mild to moderate pain** — Headache, backache, toothache, myalgia, neuralgia, pulled muscle, dysmenorrhea.
2. **Fever** — acute rheumatic fever, bacterial fever.
3. **As anti-inflammatory**: Rheumatoid arthritis, Osteoarthritis, Pericarditis, Ankylosing spondylitis.
4. **as antiplatelet**:
 - i. transient ischemic attack
 - ii. unstable angina
 - iii. Coronary artery thrombosis with MI
 - iv. Thrombosis after coronary artery surgery
5. **prevention of pre eclampsia**
6. **Helps in fetal growth**: by improving uteroplacental blood flowing in some pregnancy complication → IUG (Intrauterine growth retardation)

Indication (Bristy macam)

① To relief somatic pain (Analgesic use)

- Headache → neuralgia
- Toothache → myalgia
- Backache → Dysmenorrhea
- Post operative pain

② Treatment of fever except viral + dengue fever

③ Anti-inflammatory use:

- RA → R. Fever
- Osteoarthritis
- Brucellosis
- Tenosynovitis

④ Anti-Platelet use:

- TIA → DIC
- Acute and post MI
- Deep vein thrombosis

⚠️ ADVERSE EFFECT

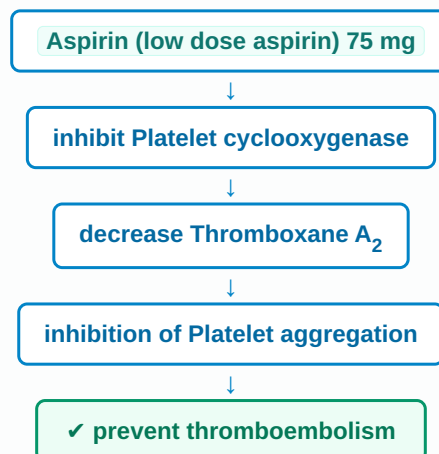
1. **GIT:** Nausea, gastric irritation, erosion, peptic ulceration, gastric bleeding or perforation, esophagitis.
2. **Renal:** 1. Na^+ , H_2O retention, 2. chronic renal failure, 3. nephropathy / papillary necrosis.
3. **CVS** — (1) rise in BP, (2) Risk of MI especially with COX-2 inhibitor.
4. **Hepatic** — (1) Raised transaminase $\uparrow$, (2) Hepatic failure.
5. **CNS** — overdose causes — Headache, confusion, vertigo, tinnitus, hearing difficulty.
6. **Allergy** — Urticaria, rhinitis, asthma exacerbation, nasal polyposis, rash, angioedema.
7. **Hematological** — Neutropenia, thrombocytopenia, Hemolytic anemia, bleeding.

✗ CONTRAINDICATION

1. Peptic ulcer
2. Bleeding disorder — haemophilia, von Willebrand disease.
3. Liver and kidney disease
4. pregnancy (1st and 3rd trimester)
5. lactation
6. History of aspirin allergy
7. pt undergoing electric surgery
8. Bronchial asthma
9. Gout
10. Dengue
11. Children and teenager having viral infection like chicken pox, flu due to developing Rey's Syndrome → encephalopathy, Liver damage.

DETAILED PHARMACOLOGICAL ACTIONS:

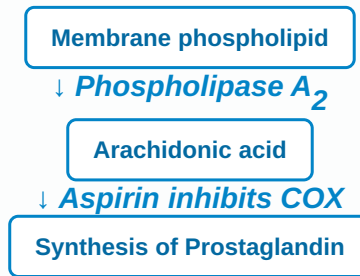
Anti Platelet Action:



Anti inflammatory Action:

next page →

Anti-inflammatory action:

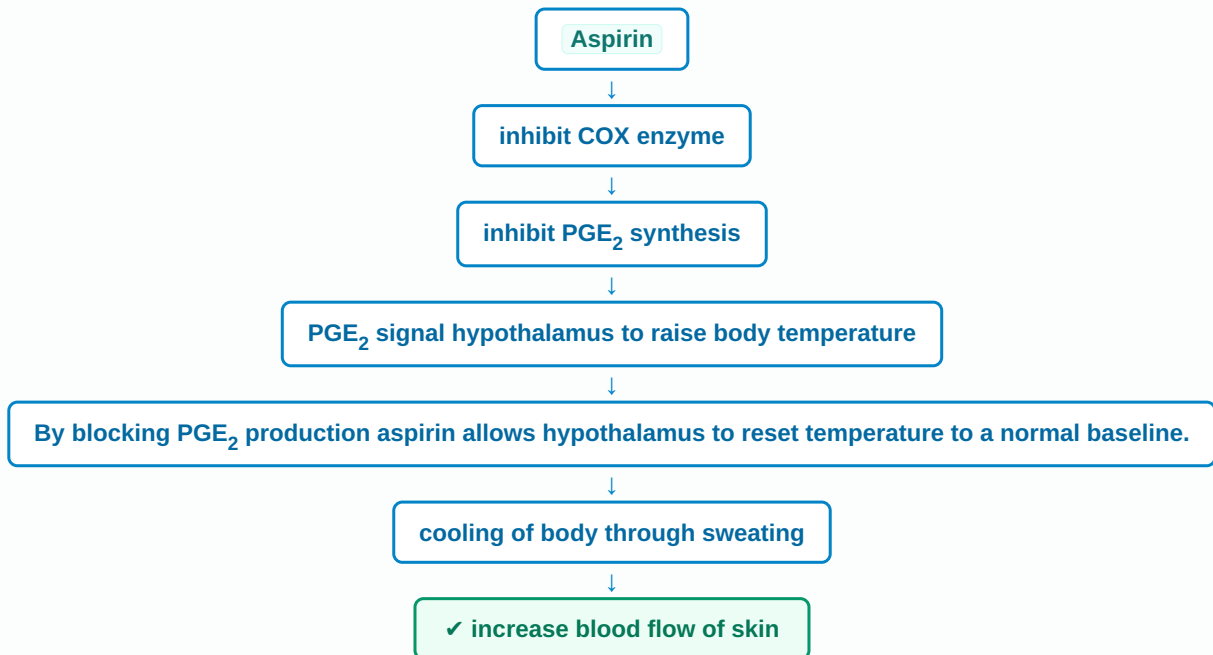


Analgesic Action:

- **Central action aspirin:** Central action aspirin → inhibit pain stimuli at Subcortical region → causes analgesia.
- **Peripheral action aspirin:** again, Peripheral action aspirin → inhibition COX → inhibit PG synthesis → no inflammation → no pain.

ANTIPYRETIC & SYSTEMIC ACTIONS OF ASPIRIN

ANTIPYRETIC ACTION:

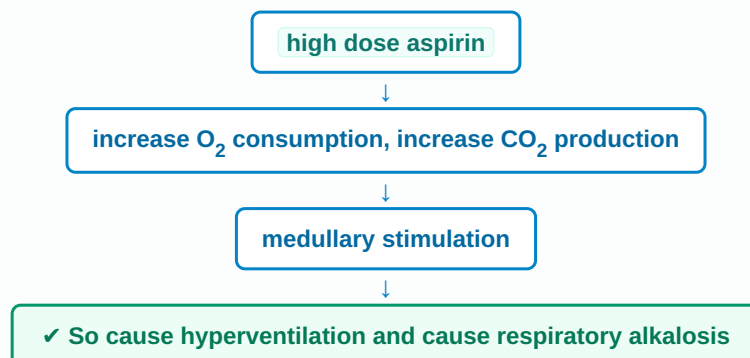


aspirin also inhibit IL-1, IL-6, which is responsible for fever (ultimately aspirin normalize body temperature)

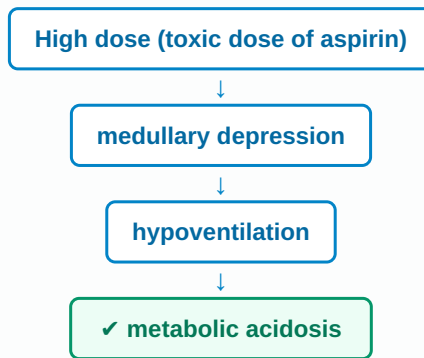
OTHERS ACTION:

(1) **Respiratory system:** 1. Respiratory alkalosis, 2. metabolic acidosis

Respiratory alkalosis:



Metabolic acidosis:



- (2) **Hematological system** — Hypoprothrombinemia
- (3) **GIT** — ulceration, bleeding and perforation.
- (4) **Liver** → 1. increase SGPT / ALT, 2. Rey's syndrome (Liver injury also encephalopathy)
- (5) **Renal effect** → (1) Hyperuricemia (low dose), (2) to salt and H₂O retention
- (6) **Endocrine effect** —
 - 1. Thyroid suppression
 - 2. Prolonged labour
 - 3. Increased steroid synthesis (change of drug interaction)

⚠ **SALICYLISM / SALICYLATE TOXICITY / SALICYLATE OVERDOSE / POISONING (ASPIRIN OVERDOSE — 8 GM)**

Early sign —

1. Tinnitus
2. Hearing loss
3. nausea, vomiting
4. Dizziness
5. Rapid breathing

Late sign:

1. High fever
2. Confusion
3. Excessive sweating (dehydrate)
4. Hallucination
5. seizure
6. cerebral edema
7. Coma

Treatment:

1. I/V fluid to combat dehydration
2. I/V infusion to NaHCO₃ to maintain high pH
3. Hemodialysis

☞ Aspirin कोमा कब तक बचता है? → Dengue + Viral fever.

⚠ **WHY ASPIRIN CONTRAINDICATED IN PREGNANCY (1ST AND 3RD TRIMESTER)**

Because:

- **In 1st trimester** — aspirin causes teratogenesis. eg. cleft lip, cleft palate.
- **In 3rd trimester** — Aspirin causes —
 - i. Delayed labour
 - ii. Profuse bleeding
 - iii. premature closure of ductus arteriosus.

PARACETAMOL (NSAID → OVER THE COUNTER DRUG)

PHARMACOLOGICAL ACTION:

1. Anti pyretic
2. Analgesic
3. weak or no inflammatory action.

✓ INDICATION

1. Fever
2. Headache
3. Toothache
4. Dysmenorrhea
5. musculoskeletal pain

⚠ ADVERSE EFFECT

1. Rash
2. chronic renal disease
3. Hepatotoxicity with acute overdose

⚠ PARACETAMOL OVERDOSE (> 10 GM → > 20 TABLET IN A SINGLE DOSE)

One of the metabolic end product of paracetamol is **N-acetyl para benzo quinoneimine (NAPQI / Quinoneimine)**. This metabolite is inactivated by conjugation with glutathione. In acute overdose, glutathione is depleted, toxic metabolites accumulate and reacts with nucleophilic constituents in cell. Thus cause liver necrosis, also involve renal tubules. → *How paracetamol causes liver damage?*

Symptoms:

- Nausea
- vomiting
- Hepatotoxicity develop after 24–48 hours of ingestion.

Management: (*written*)

1. Gastric lavage within 4 hours of ingestion.
2. Antidote — **acetylcysteine** (IV)
3. IV steroid, anti histamine to combat pseudoallergic reaction of acetylcysteine.
4. **monitoring** — PT (INR) (International normalization ratio), Serum creatinine, Blood Ph, Blood glucose.

If Patient presents after 16 hours then —

1. maintain fluid and electrolyte balance.
2. maintain blood glucose level.
3. Should be treated with fresh frozen Plasma.
4. Hemoperfusion or liver transplantation.

DICLOFENAC

✓ INDICATION

1. RA (Rheumatoid Arthritis)
2. osteoarthritis
3. Ankylosing spondylitis
4. Tendinitis (Ciprofloxacin contraindication)
5. Musculoskeletal pain
6. post operative pain
7. acute gout (**Indomethacin** best)

ANTIPYRETIC DRUG:

1. **Paracetamol**
2. **Aspirin**
3. **Indomethacin**
4. **Diclofenac**
5. **Piroxicam**

POST OPERATIVE PAIN MEDICATION:

1. **Diclofenac**
2. **ketorolac**
3. **morphine**
4. **pethidine**
5. **Ibuprofen**
6. **paracetamol**

Boistny maam (add-ons)

☐ Anti-inflammatory drugs—

① NSAIDS

② Corticosteroid
(Glucocorticoid)

③ DMARDs (Disease modifying
anti-rheumatic drug)

eg→ methotrexate,
Sulfasalazine,
Penicillamine

Extra (Bristy ma'am)

ANTI-INFLAMMATORY DRUGS & ASPIRIN ADD-ONS

Bristy Ma'am (Add-ons)

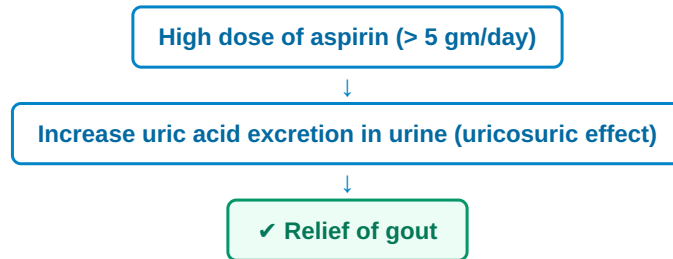
ANTI INFLAMMATORY DRUGS

1. NSAID
2. Corticosteroid (glucocorticoid)
3. Disease modifying anti rheumatic drug (DMARDs):
eg. **Methotrexate**, **Sulfasalazine**, **Penicillamine**.

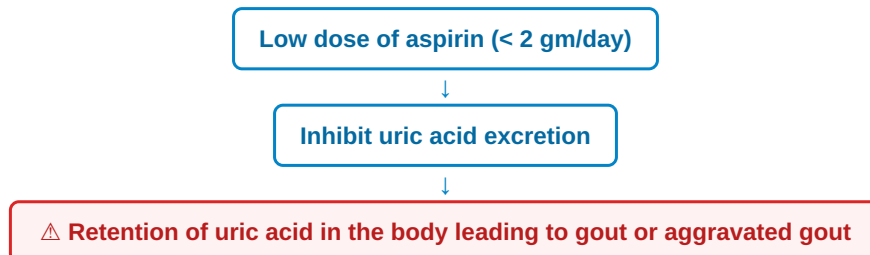
URICOSURIC EFFECT OF ASPIRIN (IMP. FOR VIVA)

i. It is dose dependent:

ii. In high dose of aspirin (over 5 gm/day):



iii. In low dose (less than 2 gm/day):

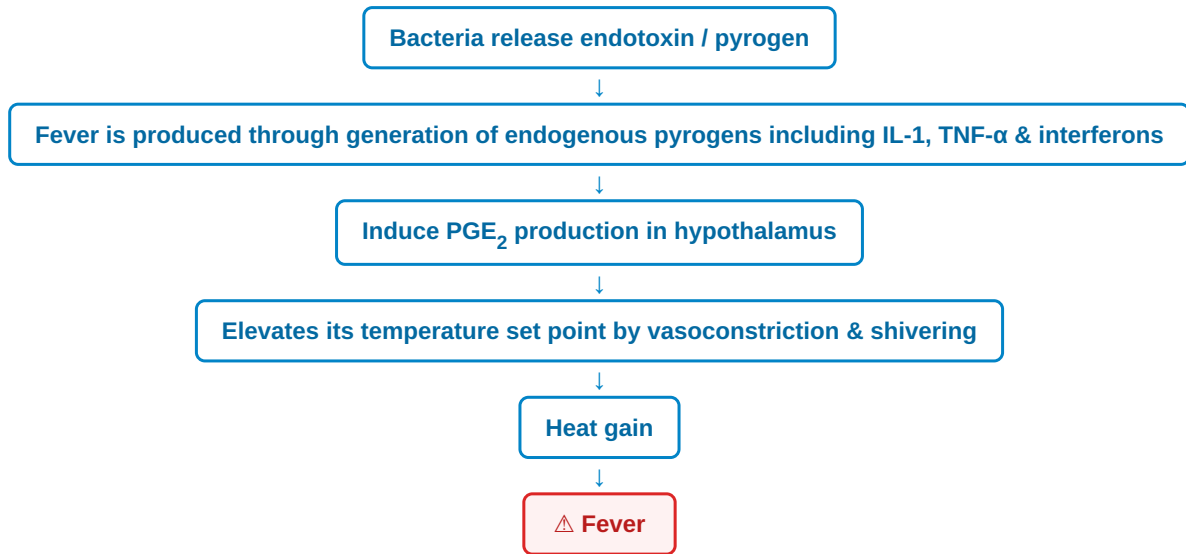


Extra (Bristy maam)

ANTIPYRETIC ACTION OF ASPIRIN

a. Normal body temperature: Maintained by a thermo-regulatory center in the hypothalamus that ensures a balance between heat production and heat loss.

b. M/A of heat / fever production:



c. M/A of antipyretic effect: Same as lecture (*blocks COX → inhibits PGE₂ synthesis → resets hypothalamic thermostat → cutaneous vasodilation & sweating → heat loss*).



OPIOID ANALGESIC: CLASSIFICATIONS

(I) ACCORDING TO SOURCE

1. Natural: morphine , codeine , dihydrocodeine
2. Semi synthetic: Diamorphine / heroine , Hydromorphone , oxycodone , hydrocodone , pholcodine , oxymorphone .
3. Synthetic: pethidine , methadone , pentazocine , fentanyl , alfentanil , remifentanil , sufentanil , tramadol (atypical)

(II) TRADITIONAL

1. Strong — morphine , diamorphine , fentanyl .
2. Intermediate —
 - a. Partial agonist: Buprenorphine , meptazinol .
 - b. Mixed agonist-antagonist: pentazocine , butorphanol , nalbuphine .
3. Weak — codeine

(III) STRUCTURAL

1. Morphinans — morphine , codeine
2. Phenylpiperidines — meperidine , fentanyl .
3. Diphenyl propylamine: eg. Methadone , dextropropoxyphene (cardiac depressant)
4. Ester — Remifentanil .

(IV) FUNCTIONAL (ON THE BASIS OF ACTIVITY ON OPIOID RECEPTOR)

1. Pure agonist — morphine , codeine .
2. Partial agonist — Buprenorphine .
3. Mixed action — Pentazocine , nalbuphine , butorphanol , nalorphine .
4. Antagonist — naloxone , naltrexone , nalmefene .

OPIOID RECEPTORS & ENDOGENOUS PEPTIDES

1. MU (M) RECEPTOR → (viva)

Location — Brainstem, Spinal cord, amygdala, nucleus accumbens, cerebral cortex, myenteric plexus in GIT

FUNCTIONS:

- i. Spinal and supraspinal analgesia
- ii. Sedative
- iii. Euphoria
- iv. miosis
- v. Respiratory depression
- vi. Slow GIT transit
- vii. muscular rigidity
- viii. physical dependence.

2. KAPPA (K) RECEPTOR

Location — Hypothalamus, spinal cord, striatum, claustrum, Peripheral nerve.

FUNCTIONS:

- i. supraspinal and spinal analgesia
- ii. Dysphoria
- iii. physical dependence.

3. DELTA (Δ) RECEPTOR

Location — Cerebral cortex, caudate putamen, nucleus accumbens, spinal cord, dorsal root ganglia, heart, parts of GIT

FUNCTIONS:

- i. supraspinal and spinal analgesia
- ii. Affective behaviour
- iii. Reduce GI motility
- iv. modulation of hormone and neurotransmitter.

ENDOGENOUS OPIOID PEPTIDE

1. β-endorphine
2. Enkephaline
3. Dynorphine

Affinity of endogenous peptide: → Read it from Table 31.1 from Katzung

MORPHINE

(Comes from Morpheus means god of dream)

✓ INDICATIONS OF MORPHINE

1. **As analgesic** — acute MI, pain due to burns, fracture, cancer pain, traumatic pain, renal colic, Postoperative pain.
2. Acute left ventricular failure
3. Acute pulmonary edema
4. preanaesthetic medication
5. **Cough** — (**codeine** , **methadone**)
6. **Diarrhoea** — (**loperamide** , **diphenoxylate**) *Traveler's diarrhoea*
7. **Colostomy** — (to increase consistency of stool)
8. **For balanced and surgical anaesthesia** — (**fentanyl** , **alfentanyl** , **Pethidine** , **sufentanil**)

⚠ ADVERSE EFFECTS OF MORPHINE

1. Respiratory depression
2. nausea, vomiting
3. Restless, hyperactivity, sedation
4. constipation
5. urinary retention
6. Bradycardia
7. Postural hypotension
8. ↑ intracranial pressure
9. Tolerance, dependence
10. Itching around nose, urticaria.

✗ CONTRAINDICATIONS OF MORPHINE

1. ^{*Severe*}undiagnosed acute abdominal pain, biliary colic, acute appendicitis (biliary pain)
2. Head injury
3. Bronchial asthma, emphysema, pulmonary fibrosis.
4. Extreme age.
5. Liver and kidney disease
6. Hypothyroidism
7. BPH
8. Hypotension, hypovolemia
9. convulsive disorder.

CNS EFFECTS & SOURCE OF MORPHINE

1. CNS INHIBITORY / DEPRESSIVE EFFECTS: *(written + viva) must*

1. Sedation
2. analgesia
3. Respiratory depression
4. cough suppression
5. Inhibition of heat regulatory center
6. dysphoria

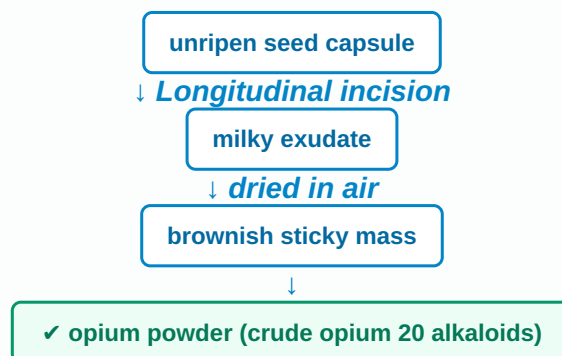
2. CNS STIMULATORY / EXCITATORY EFFECTS:

1. **Stimulation of Edinger Westphal nucleus of oculomotor nerve** → increase parasympathetic outflow, contraction of pupillary muscle → **miosis**
2. **Activation of chemoreceptor trigger zone** → causes nausea, vomiting.
3. **Hyperactive spinal cord reflex** → causes truncal rigidity
4. convulsion
5. Euphoria

Also cause — tolerance, dependance

SOURCE OF MORPHINE:

Natural source — *Papaver somniferum* → poppy flower



OPIUM CONTAINS 2 TYPES OF ALKALOID (BUT CONTAIN OTHERS):

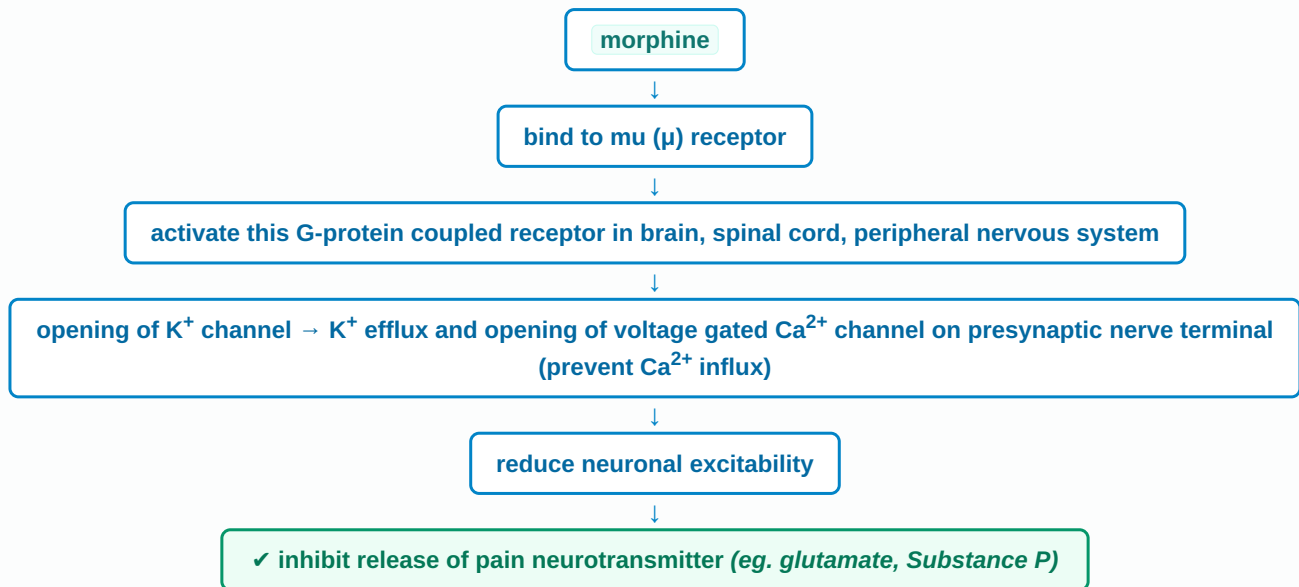
1. Phenanthrene derivatives: **morphine**, **codeine**, **Thebaine** (used as analgesic)
2. Benzoisoquinoline derivatives: **Papaverine**, **noscapine** (used as vasodilator)

DIFFERENCE BETWEEN MORPHINE AND PETHIDINE

TRAITS	MORPHINE	PETHIDINE
Source	natural	Synthetic
Potency	10 times more potent	less potent
Onset	Late	immediate
Duration of action	Longer (usually 8–10 hours)	Shorter (usually 4–6 hours)
Hypnotic effect	more	less
Respiratory Depression	more (due to histamine release)	less
Constipation and urinary retention	occur	doesn't occur
Eye	miosis	mydriasis
Histamine release	yes (cause broncho-constriction)	No
Pre anaesthetic medication	now not used	may be used
Labour	delay	not so

MORPHINE: MECHANISM & SYSTEMIC ACTIONS

MECHANISM OF ACTION:



PERIPHERAL ACTIONS OF MORPHINE:

1. GIT —

- i. produce spasm of intestinal smooth muscle.
- ii. Reduce peristalsis and gastric emptying is delayed.
- iii. Increase absorption of water
- iv. Constipation occur. (*increase tone → increase motility → decrease GIT secretion leading spasm → decrease peristalsis*)

2. Biliary tract —

- i. Constriction of biliary smooth muscle
- ii. causes biliary colic
- iii. constriction of Sphincter of Oddi
- iv. increase biliary pressure.

3. Ureter, urinary bladder:

- i. increase ureteral tone and tone in detrusor muscle of urinary bladder.
- ii. contraction of external sphincter, so cause urinary retention (*specially in post operative patient*)
- iii. decrease urinary flow (*by decrease in renal plasma flow and increase Na⁺ reabsorption*)

4. Respiratory system: Bronchospasm due to histamine release.

5. CVS — dilation of arteriole and vein, cause Postural hypotension.

Bradycardia: (due to vagal stimulation / due to stimulation of vagus nucleus)

6. Endocrine system —

- i. stimulate release of ADH, prolactin, GH, somatotropin
- ii. Inhibit release of LH, FSH

7. Uterus — prolong labour due to reduction of uterine muscle tone.

8. Miscellaneous —

- i. Flushing and warming due to cutaneous vasodilation (due to histamine release).
- ii. Itching.
- iii. Excessive sweating due to histamine release.

CLINICAL TOXICITY & MORPHINE POISONING

✗ WHY MORPHINE IS CONTRAINDICATED IN HEAD INJURY:

1. Respiratory depression and accumulation of CO_2 causes increasing of intracranial pressure and altered function of brain.
2. morphine causes miosis, vomiting and convulsion.

⚠ MORPHINE POISONING

May be accidental, suicidal or seen in drug abusers.

- 50 mg IM produce serious toxicity
- Lethal dose: 250 mg.

SIGN AND SYMPTOMS:

- | | |
|---|--|
| 1. Stupor or coma (i) | 1. hypotension (i) |
| 2. Flaccidity (ii) | 2. Shock (ii) |
| 3. Shallow and occasional breathing (iii) | 3. convulsion (iii) |
| 4. cyanosis (ix) | 4. pulmonary edema (iv) |
| 5. miosis (x) | 5. Death due to respiratory failure (viii) |

TREATMENT:

1. Gastric lavage with KMnO_4 to remove unabsorbed drug
2. I/V fluid
3. Antidote — Naloxone

1. DEFINITION & TYPES OF DEPRESSION

DEPRESSION DEFINITION

It is an **affective disorder** characterized by **sad mood, lack of interest, lack of energy**, lasting at least 2 weeks or longer.

Type of depression:

- i. Major depression (> 60%)
- ii. Bipolar affective (manic depression) (10–15)%

2. PATHOPHYSIOLOGY / CAUSE OF DEPRESSION

CAUSES OF DEPRESSION

1. Primary or idiopathic
2. Decrease amine like **NE, serotonin, Dopamine**
3. **Secondary cause:** Alcohol, **propranolol**, **α-Methyldopa**, **reserpine**, parkinsonism, MI, Left ventricular failure, vit-B₁₂, vit-B₁ deficiency.

HYPOTHESES OF DEPRESSION

1. Monoamine hypothesis: It is the main theory which states that depression is caused by functional deficit of mono-amine neurotransmitter at certain sites of brain. Monoamine neurotransmitters like Nor-adrenaline, dopamine, serotonin can be restored by antidepressant drugs. Antidepressant enhance the synaptic availability of these neurotransmitters in the brain.

• This theory is proposed by professor Schildkraut in 1965.

2. Neurotrophic hypothesis: Depression is associated with loss or fall of nerve growth factor such as **brain derived neurotrophic factor (BDNF)** which regulate neurogenesis. Effective antidepressant therapy increase neurogenesis. Though brain cells are permanent cell but in dentate gyrus, may be there are some chances of neurogenesis.

3. Neuroendocrine factor: Depression may be associated with increased Hypothalamo-pituitary-adrenal axis resulting in increase plasma Cortisol concentration, so that increase stress → Increase depression.

⑩ Treatment approaches for depression →

① Drug therapy

② Psycho-therapy includes

③ ECT (Electroconvulsive therapy)

④ CBT (Cognitive behavioral therapy)

⑤ Group therapy (to see improvement)

3. CLASSIFICATION OF ANTIDEPRESSANT DRUGS

PHARMACOLOGICAL CLASSES

1. TCA (1st Generation): (Tricyclic Antidepressant)
 - a. Noradrenaline, 5-HT reuptake inhibitor: e.g. **imipramine**, **amitriptyline**, **clomipramine**, **protriptyline**, **doxepine**, **dothiepin**.
 - b. Predominantly NA reuptake inhibitor: **desipramine**, **nortriptyline**, **reboxetine**.
2. SSRI (Selective serotonin reuptake inhibitor):
e.g. **fluoxetine**, **paroxetine**, **citalopram**, **escitalopram**, **fluvoxamine**, **dapoxetine**, **sertraline**.
3. SNRI: (Serotonin-Noradrenaline reuptake inhibitor):
e.g. **venlafaxine**, **desvenlafaxine**, **duloxetine**, **milnacipran**.
4. MAO-I:
e.g. **clorgyline**, **moclobemide**, **phenelzine**.
5. Atypical antidepressant:
e.g. **Trazodone**, **mianserine**, **mirtazapine**, **amoxapine**, **bupropion**.
6. Lithium: **Lithium**

ACCORDING TO GENERATION & SPECIAL GROUPING

According to generation:

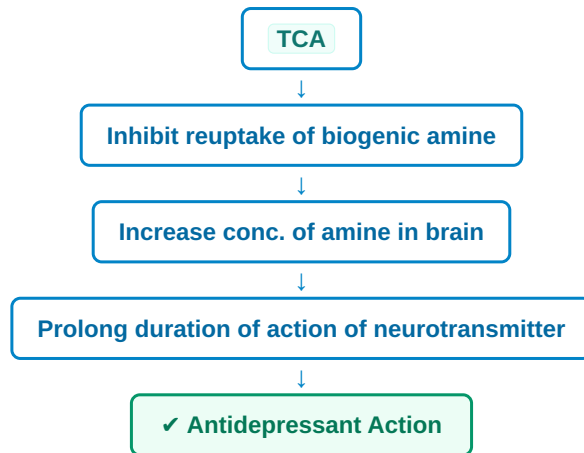
- i. 1st generation: (TCAs)
- ii. 2nd generation: e.g. **Amoxapine**, **maprotiline**
- iii. 3rd generation: e.g. **Mirtazapine**, **nefazodone**, **Venlafaxine**.

Receptor blocker: **Trazodone**, **mianserine**, **mirtazapine**

Novel compound: **Venlafaxine**, **trazodone**, **Mirtazapine**, **Reboxetine**.

4. TRICYCLIC ANTIDEPRESSANTS (TCA)

MECHANISM OF ACTION (M/A)



✓ INDICATIONS OF TCA

1. Major depression
2. Obsessive compulsive disorder (OCD)
3. Anxiety disorder
4. Neuropathic pain
5. Attention deficit disorder (ADHD)
6. Premature ejaculation
7. Enuresis (**Imipramine** used here)
8. Migraine
9. Smoking cessation
10. bulimia
11. Pruritus eg. Atopic dermatitis, lichen simplex

⚠ ADVERSE EFFECTS (A/E)

1. Sedation, confusion, light headedness.
2. **Antimuscarinic action:** DUBC (Dry mouth, Urinary retention, Blurring of vision, Constipation).
3. **Alfa adrenoceptor blocking action:** postural hypotension.
4. **H₁-receptor blocking action:** weight gain, sexual disturbances.
5. **ECG:** Prolongation of Q-T interval.
6. Tachycardia due to vagal blockage.
7. Seizure.

✗ CONTRAINDICATIONS

1. After MI
2. IHD
3. Arrhythmia
4. Pregnancy
5. Lactation
6. Following MAO-I

⚠️ ACUTE TOXICITY OF TCA

Signs & Symptoms (S/S):

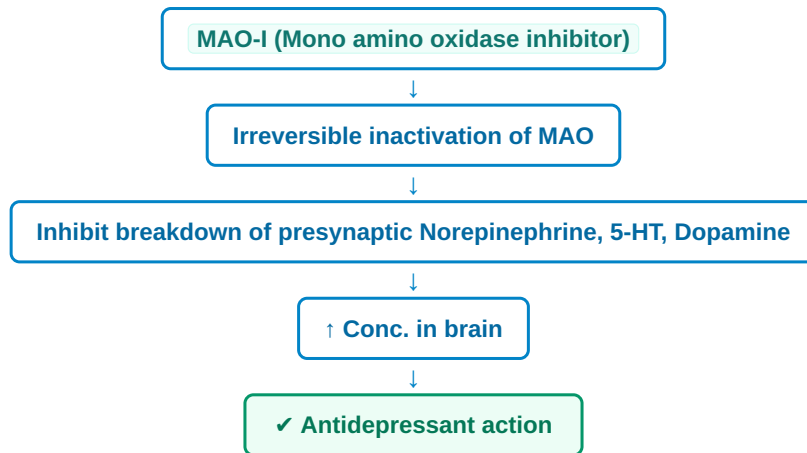
- i. Excitement
- ii. Delirium
- iii. Convulsion
- iv. Respiratory depression
- v. Dysrhythmia, Hypotension
- vi. Coma death from ventricular fibrillation.

Treatment:

- i. Gastric lavage
- ii. Activated charcoal by mouth to prevent further absorption.
- iii. I/V infusion of NaHCO_3 for correction of acidosis.
- iv. **Physostigmine** .
- v. Anti dysrhythmic drug
- vi. **Diazepam**

5. MONOAMINE OXIDASE INHIBITORS (MAO-I)

MECHANISM OF ACTION (M/A)



TYPES OF MAO-I ENZYME

1. **MAO-A:** Responsible for metabolism of NE, 5-HT
2. **MAO-B:** For dopamine metabolism

⚠ ADVERSE EFFECTS (A/E)

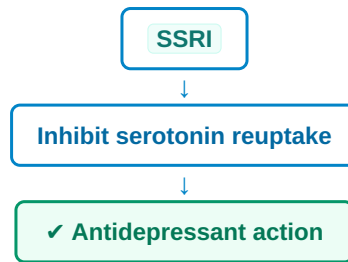
1. Excitement
2. Euphoria
3. Insomnia
4. Weight gain
5. Ankle edema
6. Postural Hypotension
7. **DUBC** → Dry mouth, Urinary retention, Blurring of vision, constipation.
8. Sexual disturbances (more)
9. **Hypertensive crisis (if taken with tyramine containing food):**
→ Throbbing Headache → BP increase → Hemorrhage → Death.

⚠ CHEESE REACTION (S/N)

Cheese, bean, chicken liver contain tyramine. Tyramine is metabolized by MAO. MAO-I prevent this. Then tyramine is absorbed and its sympathomimetic activity enhanced. Hypertensive crisis is happened (Severe throbbing headache, intracranial hemorrhage).

6. SELECTIVE SEROTONIN REUPTAKE INHIBITORS (SSRI)

MECHANISM OF ACTION (M/A)



⚠ ADVERSE DRUG EFFECTS (ADE)

1. Anorexia
2. Nausea, vomiting
3. Insomnia
4. Tremor
5. Agitation
6. Akathisia (motor restlessness)
7. Sexual dysfunction

ADVANTAGE OF SSRI OVER TCA

1. Cause no sedation
2. Cause no antimuscarinic effect
3. Well absorbed from gut
4. Does not undergo 1st pass metabolism
5. $T_{1/2}$ long (24–96) hrs.
6. Safe in pt with heart disease
7. Pt compliance is better than TCA

7. ATYPICAL ANTIDEPRESSANTS

ADVANTAGE OF ATYPICAL ANTIDEPRESSANT

1. Less side effect (Gross common effect ✗)
2. Greater therapeutic efficacy
3. Less chance of drug interaction
4. Safe in overdose
5. Effective in pt unresponsiveness to TCA, MAO-I

8. DSM-4 CRITERIA OF DEPRESSION

DIAGNOSTIC AND STATISTICAL MANUAL (DSM-4) CRITERIA

1. **Depressed mood*** (*Must ১টা থাকবেই*)
2. **Loss of interest in activities*** (*Must ১টা থাকবেই*)
3. Impaired concentration
4. Feeling of worthlessness or guilt
5. Suicidal ideation
6. Wt changes
7. Insomnia or hypersomnia
8. Loss of energy
9. Agitation

RULE: Criteria 1 or 2 must be present. Total 5 criteria must be present (যে কোন ৪টা + ১/২ এর মধ্যে ১টা = Total 5 criteria থাকবেই).

ANTI-PARKINSONISM DRUGS

PARKINSONISM DEFINITION & OVERVIEW

It is an **extra pyramidal motor disorder** characterized by **rigidity, tremor, hypokinesia** with secondary manifestations like defective posture and gait, mask-like face and sialorrhea, dementia.

If untreated, patient is rigid, unable to move, unable to breathe properly. It is a **progressive and neurodegenerative disorder**.

Another name: *Paralysis Agitans*.

CAUSES OF PARKINSONISM

1. **Idiopathic**
2. **Deficiency of dopamine in nigrostriatal pathway** and **hyperactivity of acetylcholine**. So disruption of balance between dopamine and acetylcholine.
3. **Drug induced:** **Chlorpromazine** , **reserpine** .
4. **Neurotoxin**

CLASSIFICATION OF ANTI-PARKINSONISM DRUGS

(I) DRUGS AFFECTING BRAIN DOPAMINERGIC SYSTEM

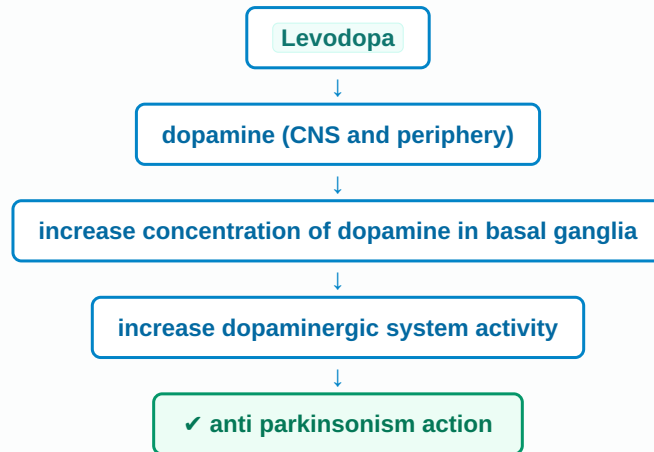
1. Dopamine precursor: Levodopa (L-dopa)
2. Peripheral decarboxylase inhibitor: Carbidopa , benserazide
3. Dopaminergic agonist: Bromocriptine , pergolide , Pramipexole , Ropinirole
4. MAO-B inhibitor: Selegiline , rasagiline
5. COMT inhibitor: Entacapone , tolcapone
6. Glutamate (NMDA) antagonist: Amantadine

(II) DRUGS AFFECTING BRAIN CHOLINERGIC SYSTEM

1. Anticholinergic: Trihexyphenidyl (Benzhexol) , Procyclidine , biperiden
2. Antihistamine: Orphenadrine , promethazine

LEVODOPA

MECHANISM OF ACTION:



PHARMACOKINETICS OF LEVODOPA

- **Route of administration:** Oral
- **Absorption:** Rapidly from small intestine.
- **Distribution:** 70% metabolized in gut, 30% enters into plasma. (1–3)% plasma concentration enters the brain unaltered.
- **$t_{1/2}$:** (1–3) hours.

COMBINATION OF LEVODOPA + CARBIDOPA (SINEMET) / DRUG INTERACTION

Levodopa is converted to dopamine by **dopa decarboxylase** present in both brain and peripheral tissue. But peripheral dopamine cannot cross the BBB (Blood-Brain Barrier).

When **levodopa** is given in combination with **carbidopa** (which does not penetrate the blood-brain barrier), the peripheral metabolism of levodopa is reduced. So more levodopa is available for entering into brain and thus increases dopamine concentration in brain.

LEVODOPA: ADVERSE EFFECTS & CLINICAL NOTES

⚠ ADVERSE EFFECTS OF LEVODOPA

1. nausea, vomiting
2. Cardiac arrhythmia including tachycardia, ventricular extrasystole.
3. Postural hypotension
4. Exacerbation of angina
5. Alteration in taste sensation
6. **Dyskinesia:** (facial tics, grimacing, tongue-thrusting, choreoathetoid movements of limbs)
7. **Behavioural effects:** Anxiety, nightmare, depression, mania, Euphoria, hallucination, confusion, Psychosis
8. **Rapid fluctuation in response (on-off phenomenon):** hypokinesia and rigidity may suddenly worsen and then improve again.
9. **Others:** mydriasis, acute glaucoma, hot flushing, brownish discolouration of saliva, elevation of BUN and liver.

SINEMET: Levodopa + Carbidopa

✗ CONTRAINDICATIONS OF LEVODOPA

1. Psychotic patient
2. Angle closure glaucoma
3. Cardiac disease
4. active peptic ulcer
5. malignant melanoma

CLINICAL JUSTIFICATIONS

Justify the use of Procyclidine in drug induced parkinsonism:

Procyclidine is an anti-muscarinic drug. It blocks the M_1 receptor in basal ganglia. So decreases the cholinergic activity. This restores balance of dopaminergic and cholinergic in CNS, so improves the tremors and rigidity.

Why Levodopa is not used in drug induced parkinsonism?

Antipsychotic drugs cause parkinsonism (drug induced parkinsonism). In psychosis, we need to reduce dopamine concentration in brain, so we use antipsychotic drugs. But later it can cause imbalance of dopamine and cholinergic balance/ratio. In that case, we cannot use levodopa because it increases the concentration of dopamine and thus increases psychosis. So anti-parkinsonism anticholinergic drugs are preferable for that kind of parkinsonism.

ANTIPSYCHOTIC DRUG (NEUROLEPTIC / ANTI-SCHIZOPHRENIC DRUG)

A. CLASSICAL ANTIPSYCHOTIC (TYPICAL) / NEUROLEPTIC DRUG

1. Phenothiazine:

- Chlorpromazine (Largactil)
- promazine
- periciazine
- trifluoperazine
- prochlorperazine (Stemetil)
- fluphenazine

→ write only drug names in exam.

2. Butyrophenone: haloperidol , benperidol

3. Substituted benzamide: sulpiride

4. Thioxanthine: flupentixol

5. Other: pimozide , loxapine

B. ATYPICAL ANTIPSYCHOTIC (NEWER)

clozapine , olanzapine , quetiapine , risperidone , ziprasidone , zotepine .

SCHIZOPHRENIA & PHARMACOKINETICS

SYMPTOM OF SCHIZOPHRENIA

Positive Symptom:

- i. hallucination
- ii. delusion
- iii. bizarre behaviour including agitation, sexual disturbances
- iv. wearing of inappropriate clothing.

Negative Symptom:

- i. **Affective flattening** manifested by unchanging facial expression with lack of communication through expression, poor eye contact, lack of responsiveness, psychomotor slowing.
- ii. **Alogia** → poverty of speech
- iii. **Anhedonia** → lack of interest & pleasure
- iv. **Apathy** → lack of energy
- v. **Attention problem.**

PHARMACOKINETIC

- **Route** — oral, parenteral, rectal
- Undergo significant first pass metabolism
- High lipid soluble, protein bound.
- **Vd** — Large
- All antipsychotic drugs are **dopamine antagonists** (activation of $D_2 \rightarrow \uparrow$ motor activity).

CAUSES OF PSYCHOSIS / SCHIZOPHRENIA

- i. dopamine receptor supersensitivity
- ii. overproduction of dopamine
- iii. Reduce dopamine destruction.

CHLORPROMAZINE

MECHANISM OF ACTION (M/A)

Chlorpromazine → Block D_2 (mainly) and D_1 receptor cause:

- i. **Antipsychotic action** — due to inhibition of dopamine receptor in **mesolimbic system**.
- ii. **Extrapyramidal action (Parkinsonism like symptoms)** — due to block of **nigrostriatal dopaminergic pathway**.
- iii. **Endocrine effect (Increase prolactin secretion)** — due to blockage of dopaminergic receptor in **tuberoinfundibular pathway**.

PHARMACOLOGICAL ACTION:

1. CNS EFFECTS

1. Apathy, aggressive tendencies reduced.
2. Antiemetic, antihiccup due to inhibition of CTZ
3. Due to effect on hypothalamus cause hypotension, hypothermia
4. Endocrine effect: e.g. amenorrhea, galactorrhea, gynaecomastia, impotence, loss of libido
5. Extrapyramidal effect: parkinsonism like symptom (bradykinesia, rigidity, tremor), tardive dyskinesia, acute dystonia, sedation

2. PERIPHERAL ACTION & CVS

- **Alfa adrenergic blocking action:** postural hypotension.
- **Antimuscarinic action (DUBC):**
 - i. **D** — Dry mouth
 - ii. **U** — Urinary retention
 - iii. **B** — Blurring of vision
 - iv. **C** — Constipation
- **CVS** — Prolong Q-T interval → ventricular arrhythmia

⊗ to prevent extrapyramidal effects →
administration of
inj. Fluphenazine
Decanoate

⊗ When we use chlorpromazine, we
also use procycline.

CHLORPROMAZINE: CLINICAL PROFILE

✓ INDICATION

1. Schizophrenia, Tourette's syndrome, ↑ depression
2. As antiemetic
3. As sedative

⚠ ADVERSE EFFECT

1. **CNS:** extrapyramidal effects like parkinsonism like symptom, tardive dyskinesia.
2. **CVS:** postural hypotension, reflex tachycardia
3. **ANS (autonomic):** DUBC
4. Sedation
5. **Endocrine effect:**
Male → Gynaecomastia, loss of libido
Female → Amenorrhea, Galactorrhea, loss of libido
6. Weight gain
7. Cholestatic jaundice
8. Rash
9. Leukopenia, agranulocytosis
10. Retinitis pigmentosa
11. Osteoporosis
12. Neuroleptic syndrome

✗ CONTRAINDICATION

1. Elderly patient with parkinsonism
2. Lactation
3. Pregnancy
4. Liver damage

ATYPICAL ANTIPSYCHOTICS & PATHWAYS

CLINICAL NOTES

- To prevent extrapyramidal effect administer inj. **fluphenazine decanoate**.
- When we use **chlorpromazine** we also use **procyclidine**.

ADVANTAGES OF ATYPICAL ANTIPSYCHOTIC

1. Better tolerated, that is less adverse effects (extrapyramidal, endocrine side effects less)
2. Effective against negative case
3. Also effective in resistant schizophrenia.

DISADVANTAGES

1. Expensive
2. Weight gain
3. ↑ Cholesterol level
4. May produce agranulocytosis with Leucopenia (**Clozapine**).

⚠ NEUROLEPTIC SYNDROME (NMS)

It is very rare & life-threatening condition while using **chlorpromazine**.

Symptoms:

- i. Fever
- ii. Confusion
- iii. Severe rigidity to muscle
- iv. Labile BP due to autonomic instability
- v. Tachycardia
- vi. Urinary incontinence or retention

DOPAMINERGIC PATHWAY

1. **Nigrostriatal pathway**: contain about 75% dopamine in the brain. Involved in coordination of voluntary movement. Associated with extra-pyramidal pathways.
2. **Mesolimbic, mesocortical pathway**: It is concerned with behavioural effects.
3. **Tuberoinfundibular pathway**: involved in control of prolactin secretion.
4. **Medullary periventricular pathway**: Involved in eating behaviour.
5. **Incertohypothalamic pathway**: Its functions are not yet defined.

(VIVA) PSYCHOTIC DISORDERS

These drugs are used to reduce psychotic problems such as:

- Schizophrenia
- Bipolar disorder
- Psychotic depression
- Senile psychosis (Aging)
- Drug induced psychosis

☒ name some psychotic disorders

⚠ (VIVA) CHLORPROMAZINE DRUG INTERACTION

Chlorpromazine induces the action of **antihistamine**, **morphine**, **pethidine**, **alcohol**, **barbiturates**, **general anesthetics**. The ultimate result is **therapeutic failure** of these drugs.

PSYCHOSIS & ANTIPSYCHOTIC DRUGS OVERVIEW

DEFINITION OF PSYCHOSIS

Psychosis: It includes a variety of mental disorders characterised by **delusion, various types of hallucination & grossly disorganized thinking.**

DOPAMINE HYPOTHESIS

↑ **Dopamine** in psychosis | ↓ **Dopamine** in depression

Excess dopaminergic activity in the **mesolimbic-mesocortical pathway** leads to abnormal psychological behaviour.

Dopaminergic overactivity causes:

- i. Receptor supersensitivity
- ii. Dopamine overproduction
- iii. Dopamine destruction reduced

✓ INDICATIONS OF ANTIPSYCHOTIC DRUGS

A. Psychotic Indications:

1. Acute schizophrenia
2. To prevent relapse of chronic schizophrenia
3. Mania
4. Acute confusional state
5. To treat anxiety (Low dose used)

B. Non-Psychotic Indications:

1. Nausea & Vomiting
2. Intractable hiccup
3. Acute intermittent porphyria

ANTI CONVULSANT DRUG

DEFINITIONS

Seizures: Episodes of brain **dysfunction** resulting from abnormal **discharge** of cerebral neurons.

It may be or may not be associated with convulsions and unconsciousness. **Motor component of seizure is called convulsion.**

Epilepsy: It is a chronic disorder characterised by **recurrent seizure**. It is due to **sudden excessive depolarization** of some cerebral neurons.

CLASSIFICATION OF SEIZURE

(i) Partial seizure:

- a. simple partial seizure
- b. complex partial seizure
- c. partial seizure secondarily generalized.

(ii) Generalized seizure:

- a. Generalized tonic clonic seizure.
- b. Absence seizure
- c. Tonic seizure
- d. Atonic seizure
- e. Clonic and myoclonic seizure
- f. Infantile spasm

CLASSIFICATION OF EPILEPSY

(i) Primary cause:

- a. 60–70% of total epilepsy
- b. Idiopathic

(ii) Secondary cause:

It is due to other disease like —

- 1. Head injury
- 2. Cerebral neoplasm
- 3. Meningitis
- 4. childhood fever / febrile convulsion.

→ Presence of convulsion
In high fever under 5 yrs
→ (Needs to be hospitalized)

PATHOGENESIS OF EPILEPSY & BASIC MODE OF ACTION

PATHOGENESIS OF EPILEPSY

In epilepsy, certain neurons or group of neurons become **hyper excitable**.

- i. These neurons begin firing burst of action potential.
- ii. That propagates in a synchronous manner to the other brain structures.
- iii. **This may be due to —**
 - Increase **cell membrane permeability** to ions particularly **voltage gated ion channel**.
 - Deficiency of **GABA activity** (Inhibitory neuron)
 - Excess activity of **excitatory neurotransmitter** like **glutamate**.

BASIC MODE OF ACTION OF ANTI CONVULSANT DRUG

1. **Reducing cell membrane permeability** to ions particularly **Na⁺ ions**.
2. **Enhancing the activity of GABA**.
3. **Inhibiting excitatory neurotransmitters** like **glutamate**.

CLASSIFICATIONS OF ANTI-EPILEPTIC DRUGS

CHEMICAL CLASSIFICATION / CLASSIFICATION OF ANTI EPILEPTIC DRUGS

1. Hydantoins — phenytoin , phosphenytoins .
2. Barbiturates — Phenobarbitone .
3. Iminostilbenes — carbamazepine , oxcarbazepine .
4. Succinimides — Ethosuximide .
5. Aliphatic carboxylic acid — valproic acid , divalproex .
6. Benzodiazepines — clonazepam , Diazepam , Lorazepam .
7. New compounds — Gabapentin , lamotrigine , topiramate , vigabatrin .

CLINICAL CLASSIFICATION — ANTI EPILEPTIC / ANTI SEIZURE DRUGS (written)

* Valproate is common across multiple seizure types.

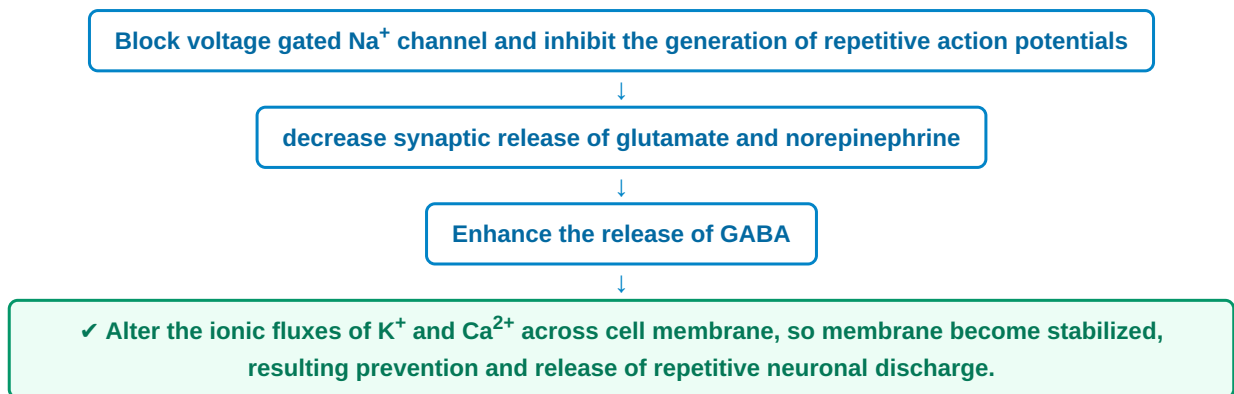
SEIZURE TYPE	DRUGS USED	NEWER DRUGS
1. Partial Seizures	phenytoin , carbamazepine , Barbiturates , * valproate	Lamotrigine , Topiramate , Zonisamide , Gabapentin , Tiagabine
2. Generalized Tonic Clonic	Phenytoin , carbamazepine , Barbiturate , * valproate	Lamotrigine , Topiramate , Zonisamide
3. Absent Seizure (petit mal)	Ethosuximide , Benzodiazepines , * valproate	Lamotrigine , Topiramate , Zonisamide
4. Others: Infantile spasm, febrile seizures, myoclonic spasm	* valproate , Benzodiazepines	Topiramate , lamotrigine , Zonisamide
5. Status epilepticus	Lorazepam , Diazepam , phenytoin , Phosphenytoin , * valproate , phenobarbitone	—

PHENYTOIN

KEY PROPERTIES & PHARMACOKINETICS

- **Oldest** anti seizure drug.
- Highly **plasma protein binding** drug.
- **Induce hepatic microsomal enzyme.**
- **Elimination of phenytoin is dose dependent:**
 - At low dose, it follows **first order kinetics**.
 - At high dose, it follows **zero order kinetics**.

MECHANISM OF ACTION



✓ INDICATIONS OF PHENYTOIN

1. partial and generalized tonic clonic seizure.
2. Status epilepticus.
3. cardiac arrhythmia

⚠ ADVERSE EFFECTS OF PHENYTOIN

1. Nystagmus, diplopia, ataxia
2. Gum hyperplasia (*To avoid, give folic acid supplement*)
3. Osteomalacia (*give vitamin-D supplement*)
4. Hirsutism
5. **Teratogenesis (fetal hydantoin syndrome):** Cleft lip, cleft palate
6. **Drug interaction:** (Drug inducer)

VALPROIC ACID

It is called **broad spectrum anticonvulsant / antiepileptic drug** because it can be used in **almost all kinds of epilepsy**.

MECHANISM OF ACTION

Block voltage gated Na^+ channel and inhibit the generation of repetitive action potentials



Inhibit GABA transaminase → Thus blocking GABA degradation



facilitate glutamic acid decarboxylase which is responsible for GABA synthesis



Inhibit GABA transport



✓ All these events causes increase concentration of GABA in the brain

✓ INDICATIONS OF VALPROIC ACID

1. Treatment:

- Absent seizure
- Generalized tonic clonic seizure (grand-mal seizure)
- Bipolar disorder

2. Prevention:

- prophylaxis of febrile convulsion.
- prophylaxis of migraine.

⚠ ADVERSE EFFECTS OF VALPROIC ACID

1. **GIT:** nausea, vomiting, diarrhoea.
2. Increase appetite and weight gain.
3. Hair loss followed by regrowth of curly hair
4. Hepatotoxicity
5. Teratogenicity (Spina bifida)

CARBAMAZEPINE

It is a **pregnancy safe anti epileptic drug**.

MECHANISM OF ACTION

1. Block voltage gated Na^+ channel and inhibit the generation of repetitive action potentials.
2. ↓ Synaptic release of glutamate.
3. potentiate the post synaptic action of GABA (Enhance).

✓ INDICATIONS OF CARBAMAZEPINE

1. Generalized tonic clonic seizure (grand-mal seizure)
2. Bipolar disorder
3. Trigeminal neuralgia

⚠ ADVERSE EFFECTS OF CARBAMAZEPINE

1. **GIT** → Nausea, vomiting
2. Diplopia, ataxia
3. **Aplastic anemia** (*anemia occurring with antiepileptic drug*)
4. Agranulocytosis

STATUS EPILEPTICUS & MANAGEMENT

DEFINITION & CLINICAL OVERVIEW

It occurs at **prolonged seizure** or a **series of seizures**, convulsive or non convulsive **without full recovery between the episodes**.

Generalized tonic seizure is most common.

It is a **life threatening condition**. Requires **immediate cardiovascular, respiratory, metabolic, pharmacological management**.

MANAGEMENT OF STATUS EPILEPTICUS

(1) Early Treatment:

- **Lorazepam** 4 mg I.V, repeat **once** after **10 min** if necessary.
- OR, **clonazepam** 1 mg I.V over 30 sec, repeat if necessary (slow infusion).
- OR, **diazepam** 10–20 mg I.V, over 2–4 min, repeat once after 30 min if necessary.

If doesn't respond, then — Established Management:

1. **Phenytoin** (15–18 mg/kg) I.V, at a rate of **50 mg/min**.
2. **Phenobarbital** (10–20 mg/kg) I.V, at a rate of **100 mg/min**.

ANESTHESIA: GENERAL OVERVIEW

DEFINITIONS

Anesthesia: Anesthesia means partial or complete loss of sensation with or without loss of consciousness.

It is of 2 types:

- i. **General anesthesia (G/A):** Loss of consciousness.
- ii. **Local anesthesia:** Without loss of consciousness.

General Anesthesia (G/A) Definition: It is a condition characterized by reversible and controllable loss of consciousness, analgesia, amnesia (loss of memory), inhibition of reflexes, and skeletal muscle relaxation.

General Anesthetics: Drugs that produce reversible and controllable loss of consciousness, analgesia, amnesia, inhibition of reflexes, and skeletal muscle relaxation are called general anesthetics.

PROPERTIES OF AN IDEAL GENERAL ANESTHETIC

1. Produce rapid, smooth, and pleasant induction.
2. Produce controllable and rapidly reversible loss of consciousness.
3. Allow pleasant and rapid recovery after discontinuation.
4. Have a wide range of safety margin (Low Therapeutic Index / TI).
5. Be a non-irritant, non-inflammable, and non-explosive agent.
6. Produce minimal adverse effects (A/E).
7. Not produce capillary bleeding.
8. Not have significant undesirable interaction with other drugs.

CLASSIFICATION OF GENERAL ANESTHETIC AGENTS

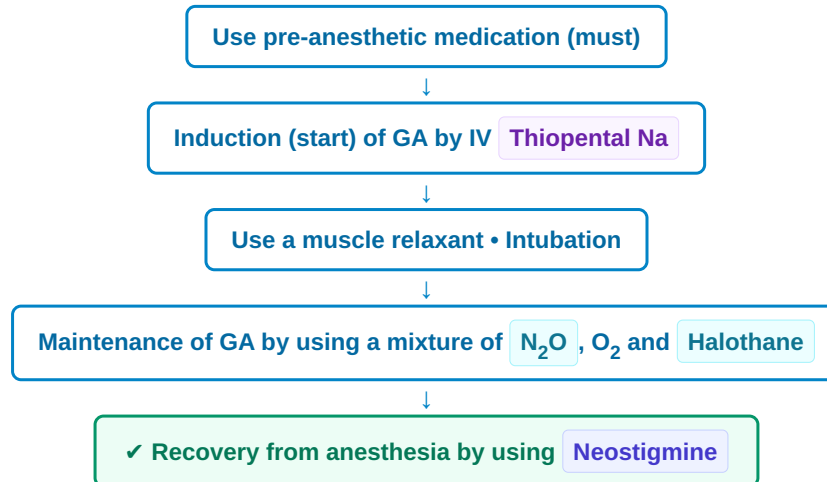
1. INHALATIONAL	2. PARENTERAL (IV)
Gas: Nitrous Oxide Volatile Liquids: Halothane , Isoflurane , Desflurane , Sevoflurane , Enflurane	Barbiturates (Ultrashort acting): Thiopental Na Opioids: Fentanyl , Alfentanil Benzodiazepines: Midazolam , Diazepam Others: Etomidate , Propofol (milky white), Ketamine

ANESTHETIC PROTOCOLS & STAGES

BALANCED ANESTHESIA

Balanced anesthesia refers to the use of multiple drugs to achieve desired clinical outcomes, such as unconsciousness, immobility, muscle relaxation, and control of vital signs, during surgery.

Protocol for long duration operations:



Note: N₂O, O₂ and Thiopental Na combination is used to get a synergistic effect. All these procedures collectively (from pre-anesthetic medication to recovery) are known as balanced anesthesia.

BASAL ANESTHESIA

It is the anesthesia which is produced by parenteral administration of one or more drugs to form the base of GA (used to start the anesthesia process).

Drugs used for basal anesthesia: Thiopental Na, Propofol → Milky white colour.

STAGES OF ANESTHESIA

- ✓ 1. **Stage I: Stage of Analgesia** — Analgesia without amnesia.
- ✓ 2. **Stage II: Stage of Excitement** — Nausea, vomiting, hyperreactivity, irregular respiration.
- ✓ 3. **Stage III: Stage of Surgical Anesthesia (Operation is performed in this stage)** — Sleep, normal respiration, and normal blood pressure.
- ✓ 4. **Stage IV: Medullary Depression / Paralysis** — Depression of vasomotor and respiratory centers leading to coma and death.

PRE-ANESTHETIC MEDICATION ** Must

Definition: The use of drugs prior to induction of GA to make it more pleasant and safe is known as pre-anesthetic medication. (Usually given 20 minutes to 3 hours pre-operatively).

DRUG CLASS / DRUG	PURPOSE
1. Benzodiazepines (Diazepam , Lorazepam , Midazolam)	To produce sedation and reduce anxiety and stress.
2. Opioids (Morphine , Pethidine)	Used if preoperative pain is present and to produce good analgesia.
3. Anticholinergics (Atropine)	To compensate salivation, gastric, and bronchial secretions, and counter bradycardia.
4. H ₂ Blockers / PPI	To prevent gastric regurgitation and aspiration pneumonia.
5. Antiemetics (Metoclopramide , Domperidone)	To reduce vomiting and gastroesophageal reflux.
6. Antihistamines (Promethazine)	To produce sedation and potentiate analgesic effect.

GENERAL ANESTHETIC PHARMACOLOGY

MECHANISM OF ACTION OF GENERAL ANESTHETICS (GA)

1. **Old Theory:** The anesthetic agent binds with the lipid matrix of the nerve membrane. This leads to secondary changes in ion flux and interaction with membranes of ligand-gated ion channels.
2. **GABA Facilitation:** Facilitation of GABA receptor function.
3. **Glutamate Antagonism:** Antagonizes the action of the excitatory neurotransmitter glutamic acid (glutamate) at N-methyl-D-aspartate (NMDA) receptors.

THIOPENTAL SODIUM (THIOPENTAL NA)

Thiopental is an **ultra-short-acting barbiturate**.

Indications: Used to **induce general anesthesia**, **treat convulsions** and **reduce intracranial pressure**.

PHARMACOKINETICS & REDISTRIBUTION:

- **Thiopental Na** is a very fast-acting anesthetic with an onset of action within 30–40 seconds.
- One IV bolus provides anesthesia for 5–10 minutes.
- Redistribution occurs quickly (within 2–3 minutes). It is a highly lipid-soluble compound. Its short duration of action is due to redistribution into muscle and eventually fat.

Advantages:

1. Potent anesthetic.
2. Rapid induction and pleasant recovery.
3. Non-irritant and non-explosive.
4. No hepatotoxicity and no nephrotoxicity.
5. Post-operative complications are less.

Disadvantages:

1. Short period of anesthesia (maximum 10 minutes).
2. Poor analgesia.
3. May produce apnea.
4. Depresses vasomotor center and respiratory center.
5. May cause circulatory collapse.

DISSOCIATIVE ANESTHESIA / KETAMINE

Dissociative anesthesia usually lasts for about 10 minutes. It is characterized by **catatonia, amnesia, and analgesia without actual loss of consciousness**. **Ketamine** is used for dissociative anesthesia.

INDICATIONS FOR KETAMINE:

1. Burn dressing (particularly in children).
2. Short duration surgical procedures (e.g., circumcision).
3. Short diagnostic procedures.
4. In patients with cardiovascular shock.

Note: **Ketamine** is the only IV anesthetic that causes an increase in heart rate, cardiac output, and blood pressure. It also has bronchodilation and analgesic effects.

ADVERSE EFFECTS OF GENERAL ANESTHETICS

⚠ INTRA-OPERATIVE ADVERSE EFFECTS (DURING SURGERY)

*Normally anesthetics decrease BP and HR, except **Ketamine**.*

1. Respiratory depression (in overdosage).
2. Salivation and bronchial secretions.
3. Cardiac arrhythmia.
4. Fall of blood pressure (Hypotension).
5. Aspiration of gastric contents.
6. Laryngospasm and asphyxia.

⚠ POST-OPERATIVE ADVERSE EFFECTS (AFTER SURGERY)

1. Nausea and vomiting.
2. Persisting sedation.
3. Pneumonia (secondary to aspiration).
4. Organ toxicities resulting in liver and kidney damage (evaluated via RME, Liver Function Tests).
5. Nerve palsy (related to spinal anesthesia).
6. Delirium.
7. Cognitive defects (in elderly patients following long duration anesthesia).

LOCAL ANESTHETICS (LA)

DEFINITION

A local anesthetic can be defined as a drug which reversibly prevents transmission of nerve impulses in the region to which it is applied, without affecting consciousness.

CLASSIFICATION OF LOCAL ANESTHETICS

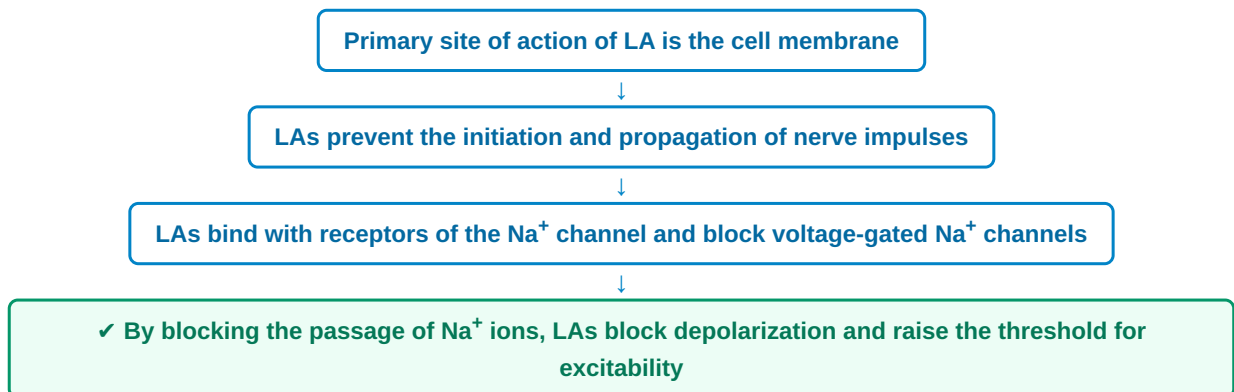
A. According to Route of Administration:

1. Surface Anesthesia: Lidocaine, Tetracaine, Benzocaine.
2. Infiltration Anesthesia: Lidocaine, Bupivacaine, Procaine.
3. Intravenous Regional Anesthesia: Lidocaine, Prilocaine.
4. Nerve Block Anesthesia: Lidocaine, Bupivacaine.
5. Spinal Anesthesia: Lidocaine, Bupivacaine, Tetracaine. (Anesthetizes the area under the spinal cord; upper body parts remain normal).
6. Epidural Anesthesia: Lidocaine, Bupivacaine. (Used in normal delivery to reduce pain).

B. According to Duration of Action:

- Short Acting: Procaine
- Intermediate Acting: Lidocaine
- Long Acting: Tetracaine, Bupivacaine

MECHANISM OF ACTION OF LOCAL ANESTHETICS



LOCAL ANESTHETICS: CLINICAL PROFILE

⚠ ADVERSE EFFECTS (A/E) OF LOCAL ANESTHETICS

1. Immediate Effects:

- a. **Hypersensitivity:** Anaphylactic shock, angioedema.
- b. **CNS Stimulation:** Anxiety, excitement, euphoria, salivation, tremor, headache.
- c. **Neurological / Sensory:** Hearing disturbances, slurring of speech.
- d. **CNS Depression (following stimulation):** Fall of BP, respiratory depression, coma.
- e. **Technical Faults:** Pain, broken needle, injury to veins.

2. Late Effects (Post-operative):

- a. Ischemia of localized area.
- b. Trismus (difficulty in opening mouth).
- c. Infection, ulceration.
- d. Facial palsy.
- e. Impairment of vision.

✓ INDICATIONS OF LOCAL ANESTHETICS

1. **Minor Surgeries:** Hernia repair, Vasectomy, Tubectomy, Cesarean section.
2. Extraction of tooth.
3. Incision and drainage of localized infections.
4. Episiotomy, removal of cysts.
5. Topical application as spray, jelly, or ointment to relieve pain.

PROLONGATION OF DURATION OF ACTION OF LOCAL ANESTHETICS

The duration of action of LA is prolonged by:

1. Increasing the dose.
2. Adding a vasoconstrictor agent (e.g., **Adrenaline**).

Role of Adrenaline with LA:

Adrenaline causes local vasoconstriction at the injection site → leads to sustained and delayed absorption of LA agents → results in prolongation of anesthetic action and reduced local bleeding.

DRUGS USED IN MALARIA

Malaria occurs due to the biting of female *Anopheles* mosquitoes. → infected by *plasmodium* species.

1. CLASSIFICATION OF ANTIMALARIAL DRUGS

A. For Clinical Attack (Blood Schizonticide):

Drugs that act on erythrocytic parasites are called schizonticides.

Chloroquine, Quinine, Mefloquine, Artesunate, Artemether, Artemisinin, Pyrimethamine

B. For Radical Cure (Tissue Schizonticide, Causal Prophylaxis):

Drugs that eliminate developing or dormant stages of parasites (hypnozoites) in liver tissue.

Primaquine, Proguanil

C. Chemoprophylaxis:

Chloroquine, Mefloquine, Atovaquone, Proguanil

D. Anti-Relapse:

Drugs that harm sexual stages and prevent transmission to mosquitoes.

Primaquine, Proguanil

E. Gametocidal:

Drugs that kill sexual stages and prevent transmission to mosquitoes.

Primaquine, Chloroquine

Plasmodium species →

① *P. vivax*

② *P. Malariae*

③ *P. ovale*

④ *P. Falciparum*

Intermediate host → Human

Definitive host → Mosquito

SPECIAL CONSIDERATIONS & RADICAL CURE

2. SPECIAL CONSIDERATIONS

Drugs Safe in Pregnancy:

- Chloroquine
- Mefloquine (Unsafe in 1st trimester)

Combinations:

- Fansidar: Sulfadoxine 500mg + Pyrimethamine 25mg
- Malarone: Atovaquone 250mg + Proguanil 100mg
- Coartem: Artemether 20mg + Lumefantrine 120mg

3. AIM OF ANTI-MALARIAL DRUG THERAPY → written

1. To prevent and treat clinical attack of malaria.
2. To completely eradicate the parasite from the patient.
3. To reduce the human reservoir of infection.

4. RADICAL CURE (FOR PLASMODIUM VIVAX AND OVALE)

Complete elimination of malarial parasite from the body, specifically the elimination of **dormant liver hypnozoites**.

Treatment: Primaquine (15mg daily for 14 days after travel, up to 21 days).

DRUG PROFILE: CHLOROQUINE

PROPERTIES & PHARMACOLOGICAL ACTIONS

Properties:

- Synthetic 4-aminoquinoline.
- Highly effective blood schizonticide.
- Has a large volume of distribution (13,000 L).
- Has no effect on exo-erythrocytic stages.

Pharmacological Action:

- i. Anti-malarial action
- ii. Anti-inflammatory action
- iii. Amoebicidal

Pharmacokinetics:

- Plasma $T_{1/2}$: Long (5day)
- Excretion: Urine (kidney)

MECHANISM OF ACTION (MOA)

1. **Inhibition of plasmodium nucleic acid synthesis:** Chloroquine blocks the synthesis of DNA and RNA in plasmodial cells by forming a complex with DNA. It prevents replication or transcription to RNA.
2. **Inhibition of digestion in the parasite:** Chloroquine is taken up by parasitized RBCs → Drug concentrates in lysosomal vacuoles → Alteration of lysosome pH → Interferes with lysosomal breakdown of hemoglobin. It also interferes with phospholipid metabolism within the parasite.

✓ INDICATIONS OF CHLOROQUINE

1. Malaria (for treatment and prophylaxis)
2. Rheumatoid Arthritis (RA)
3. Systemic Lupus Erythematosus (SLE), Discoid Lupus Erythematosus
4. Amoebic liver abscess

✗ CONTRAINDICATIONS OF CHLOROQUINE

- Hepatic/Liver disease
- Retinal and visual field abnormalities
- Blood disorders
- Psoriasis

⚠ ADVERSE DRUG EFFECTS (ADE) OF CHLOROQUINE

Normal Dose:

- Nausea, vomiting, abdominal discomfort
- Headache, itching, fatigue, tinnitus
- Hemolysis

High Dose:

- Retinopathy (due to toxic accumulation in retina)
- Defect in color vision (Blue-black)
- Photophobia
- Subcapsular cataract
- Ototoxicity
- Myopathy
- Peripheral neuropathy

QUININE, TOXICITY & CHEMOPROPHYLAXIS

DRUG PROFILE: QUININE

Source: Cinchona Bark • **Half-life ($T_{1/2}$):** 11 hours

Mechanism of Action (MOA): Quinine binds with plasmodial DNA to inhibit protein synthesis.

INDICATIONS:

- Treatment of severe falciparum malaria.
- Treatment of chloroquine-resistant malaria.
- Night leg cramp.

ADVERSE DRUG EFFECTS (ADE):

- i. Nausea, vomiting, epigastric pain.
- ii. Cinchonism (headache, nausea, dizziness, tinnitus, visual disturbance, mental confusion).
?! due to repeated admin. of quinine within therapeutic dose.
- iii. Hemolysis.
- iv. Hypoglycemia.

⚠ 5. OTHER ADVERSE DRUG EFFECTS

- **Pyrimethamine:** Megaloblastic anemia
- **Primaquine:** Methemoglobinemia / Hemolysis

6. CHEMOPROPHYLAXIS REGIMENS

Chloroquine-Sensitive Area: *for travellers.*

Chloroquine 150mg (2 tabs weekly) OR **Proguanil 100mg** (1–2 tabs daily). Start 2–3 days before travel, up to 4 weeks.

Chloroquine-Resistant Area:

Mefloquine 250mg (1 tab weekly) OR **Doxycycline 100mg** daily OR **Malarone** (1 tab daily).

7. TREATMENT OF MALARIA (CLINICAL SETTINGS)

1. CHLOROQUINE SENSITIVE (*P. FALCIPARUM*, *P. MALARIAE*)

Drug Therapy: **Chloroquine 1g**, followed by 500mg at 6 hours, 24 hours, and 48 hours.

2. CHLOROQUINE SENSITIVE (*P. VIVAX*, *P. OVALE*)

Drug Therapy: **Chloroquine** followed by 500mg at 6 hours, 24 hours, and 48 hours. After the chloroquine course is completed, give **Primaquine** for 14 days.

3. UNCOMPLICATED MALARIA WITH CHLOROQUINE-RESISTANT *P. FALCIPARUM*

Drug Therapy: **Coartem** (4 tabs BD for 3 days) **OR** **Malarone** (1 tab daily for 3 days) **OR** **Mefloquine 750mg** followed by 500mg 6–8 hourly for 3 days + **Quinine Sulphate** for 3 days.

4. SEVERE OR COMPLICATED *P. FALCIPARUM*

Drug Therapy Options:

- **Artesunate 2.4 mg/kg** IV 12-hourly for 2 days, followed by **Doxycycline 100mg** for 7 days, then **Coartem** or **Malarone** or **Mefloquine** for 3 days; **OR**
- **Artemether 3.2 mg/kg** IM 12-hourly; **OR**
- **Quinine dihydrochloride 20 mg/kg** IV infusion, then 10 mg/kg 8-hourly; **OR**
- **Quinine Sulphate** + **Doxycycline** (Adult) / **Clindamycin** (Child) for 7 days; **OR**
- **Quinidine gluconate**.

8. MALARIA: RELAPSE AND RECRUDESCENCE

RELAPSE	RECRUDESCENCE
Definition: Occurs when dormant liver stage parasite (hypnozoite) reactivates after initial infection, causing a new episode of disease. Species: Usually caused by <i>P. vivax</i> and <i>P. ovale</i>. Timeline: 3 weeks to 3 years after initial illness, depending on the geographic strain of the parasite. Treatment: Primaquine for 2–3 weeks.	Definition: Incomplete clearing of the parasite from RBCs during treatment, meaning treatment failure. Occurs due to drug resistance, wrong dose, or missing drugs. Species: Can happen with any species of malaria parasite. Timeline: Usually happens within the first 2–4 weeks of treatment. Treatment: Artemether + Lumefantrine for 5 days OR Malarone OR Quinine with Tetracycline / Doxycycline / Clindamycin for 7 days.

KALA-AZAR (VISCERAL LEISHMANIASIS)

ETIOLOGY

Causative agent: *Leishmania donovani*

TREATMENT REGIMENS

Primary Treatment:

- **Na⁺ stibogluconate / sodium antimony gluconate (SAG)** 20 mg/kg/day I/V or I/M for 28 days
OR
- **Miltefosine** 2.5 mg/kg/day for 28 days

Treatment Failure Case:

- **Pentamidine** 4 mg/kg I/M 3 times weekly for 5 weeks
- On 6th week, SAG course starts and continues for 30 days.

PKDL (Post Kala-Azar Dermal Leishmaniasis):

- **SAG** 20 mg/kg/day for 120 days, with 10 days interval per 120 days.

⚠ ADVERSE EFFECTS OF NA⁺ STIBOGLUCONATE (SAG)

1. Nausea, vomiting
2. Fever, headache, rash
3. Myalgias & arthralgia
4. Abscess
5. Arrhythmia
6. Hemolytic anemia
7. Renal-Liver toxicity

⚠ ADVERSE EFFECTS OF PENTAMIDINE

1. **Hypotension, tachycardia, dizziness, dyspnea** → due to rapid I/V infusion
2. **Sterile abscess** at injection site due to I/M administration
3. Pancreatic toxicity
4. Renal insufficiency (reversible)
5. Cough, dyspnea
6. **Others:** rash, fever, nausea
7. Vomiting, arrhythmia, hallucination, hypocalcemia, thrombocytopenia